

Unit 16

HYDROXY COMPOUNDS

63 items · 204 marks

Syllabus sections in this unit

16.1 Alcohols

Unit 16: Hydroxy compounds

1. M/J 26/P21/Q4/a(i),b(i,iii)

- 4 (a) Compound **Z** is oxidised to produce organic compound **W**, $\text{C}_5\text{H}_6\text{O}_7$, by heating under reflux with an excess of acidified $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$.

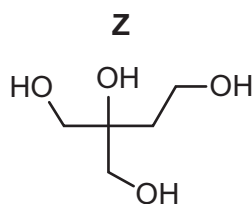


Figure 4.1

- (i) Describe the colour change in this reaction.

colour of mixture at start of the reaction

colour of mixture after the reaction

[1]

- (b) Compound **A** is a structural isomer of **Z**.

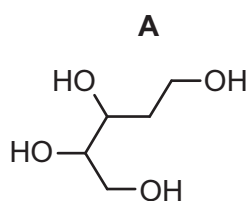


Figure 4.2

- (i) Complete Table 4.1 to show what is observed when separate samples of **A** are added to each reagent. State 'none' if no reaction occurs.

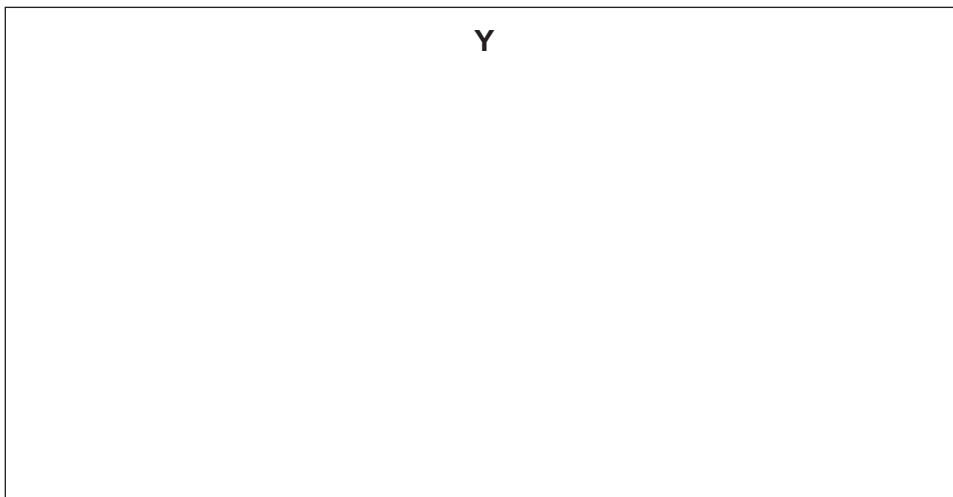
Table 4.1

reagent	observation with A
Na(s)	
2,4-dinitrophenylhydrazine (2,4-DNPH)	
Fehling's reagent	none

[2]

- (iii) Compound **A** is heated under reflux with an excess of acidified $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$ to produce organic compound **Y**.

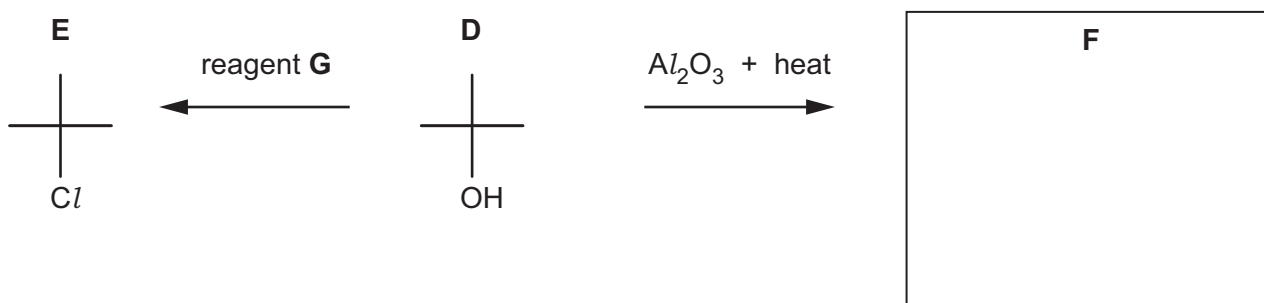
Draw the structure of **Y**.



[2]

2. M/J 26/P22/Q4/a,b(i)

4 (a) Figure 4.1 shows two reactions of compound **D**.

**Figure 4.1**

(i) **D** reacts with a single reagent, **G**, to produce compound **E**. Identify reagent **G**.

[1]

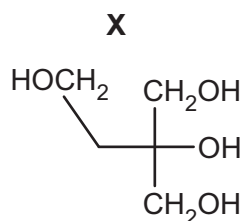
(ii) **D** is heated with Al_2O_3 to produce organic compound **F**.
Identify the role of Al_2O_3 in this reaction.

[1]

(iii) Draw the structure of **F** in Figure 4.1.

[1]

(b) The structure of compound **X** is shown in Figure 4.2.

**Figure 4.2**

The molecular formula of **X** is $\text{C}_5\text{H}_{12}\text{O}_4$. Complete oxidation of 0.255 g of **X** occurs when it reacts with exactly 10.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$.

(i) Describe the colour change in this reaction.

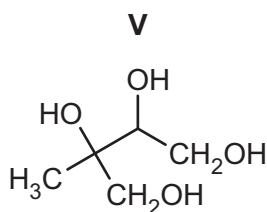
colour of mixture at start of the reaction

colour of mixture after the reaction

[1]

3. M/J 26/P23/Q4/a(i,ii),b,c(iii),d(i)

4 The structure of compound **V** is shown in Figure 4.1.

**Figure 4.1**

(a) (i) Complete Table 4.1.

Table 4.1

	primary hydroxyl groups	secondary hydroxyl groups	tertiary hydroxyl groups
number of groups in one molecule of V			

[2]

- (ii) Complete Table 4.2 to show what is observed when separate samples of **V** are added to each reagent. State 'none' if no reaction occurs.

Table 4.2

reagent	observation with V
Na(s)	
2,4-dinitrophenylhydrazine (2,4-DNPH)	
Fehling's reagent	none

[2]

- (b) Figure 4.1 is repeated for use in 4(b).

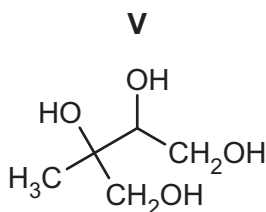


Figure 4.1

V is completely oxidised by heating under reflux with an excess of acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ to produce organic compound **Z**.

- (i) Describe the colour change in this reaction.

colour of mixture at start of the reaction

colour of mixture after the reaction

[1]

- (ii) Draw the structure of **Z**.

[1]

- (c) Compounds **W**, **X** and **Y** are structural isomers of **V**.

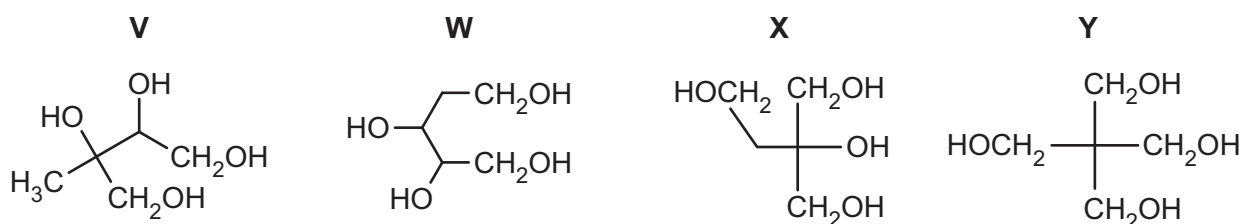


Figure 4.2

(iii) **Y** reacts with reagent **R** to produce compound **E**.

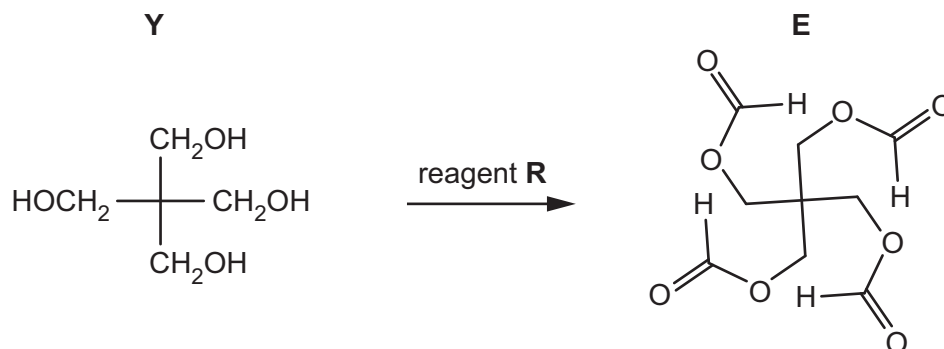


Figure 4.3

Name reagent **R**.

[1]

(d) Exactly 0.03 mol of one of the isomers, **V**, **W**, **X** or **Y**, is completely oxidised by 0.08 mol of acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ to produce only **one** organic compound with molecular formula $\text{C}_5\text{H}_4\text{O}_8$.

(i) Identify which isomer, **V**, **W**, **X** or **Y**, is used in this reaction.

[1]

4. M/J 26/P24/Q4/a,b

- 4** (a) Alcohol **S** has the structural formula $\text{CH}_3\text{C}(\text{OH})(\text{CH}_3)\text{CH}_2\text{CH}_3$.

Classify alcohol **S**.

[1]

- (b) Figure 4.1 shows compound **T**.

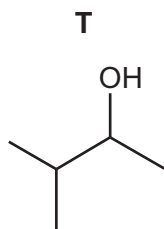


Figure 4.1

Tests are carried out on **T** using different reagents.

Complete Table 4.1 to show the observations of each test on **T**. If no reaction occurs, write 'no reaction' in the relevant box.

Table 4.1

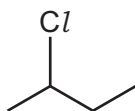
reagent	observations
Na	
acidified $\text{K}_2\text{Cr}_2\text{O}_7$	
alkaline $\text{I}_2(\text{aq})$	

[3]

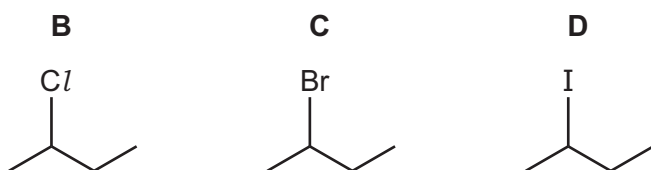
5. MAR 26/P22/Q3/e

3 Secondary halogenoalkanes are important building blocks in organic chemistry.

2-chlorobutane is an example of a secondary halogenoalkane.

2-chlorobutane**Figure 3.1**

(d) The rates of hydrolysis of three secondary halogenoalkanes, **B**, **C** and **D**, are compared.

**Figure 3.3**

Each halogenoalkane is dissolved in ethanol and reacted with $\text{AgNO}_3(\text{aq})$.

Identify the order of the rates of hydrolysis of compounds **B**, **C** and **D**. Explain your answer.

fastest
>
>
slowest

explanation

[2]

(e) The organic product of hydrolysis of each of **B**, **C** and **D** is butan-2-ol.

Complete Table 3.1 to give details of some typical chemical reactions of butan-2-ol.

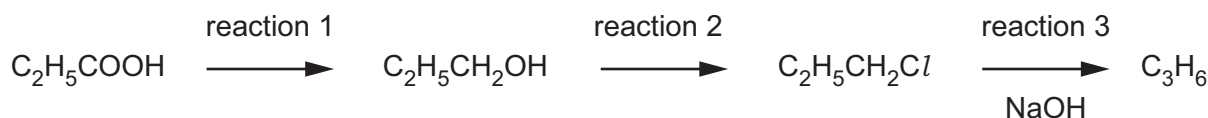
Table 3.1

reagent and conditions	observation with butan-2-ol	organic product(s)
acidified $\text{K}_2\text{Cr}_2\text{O}_7$		
Na(s)	effervescence	
	yellow precipitate	and CHI_3

[3]

6. O/N 25/P21/Q4/a

also Unit 15 (2 marks), Unit 18 (1 mark); this unit 2 marks

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .**Fig. 4.1****(a) (i)** Identify the type of reaction that occurs in reaction 1.

[1]

(ii) Suggest a suitable reagent for reaction 2.

[1]

(iii) Reaction 3 is an elimination reaction.

Write an equation for this reaction and identify the solvent and conditions used.

equation

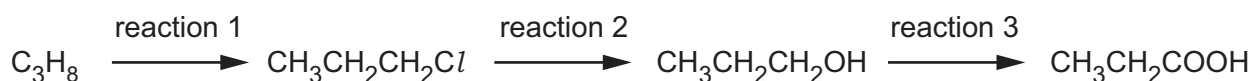
solvent and conditions

[2]

(iv) $C_2H_5CH_2OH$ can be directly converted to C_3H_6 .

Suggest the reagent and conditions for this conversion.

[1]

7. O/N 25/P22/Q4/c**4** Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, CH_3CH_2COOH .**Fig. 4.1****(c)** Reaction 3 takes place when $CH_3CH_2CH_2OH$ is heated under reflux with acidified potassium dichromate(VI) solution.**(i)** State the colour change that takes place in the reaction mixture.

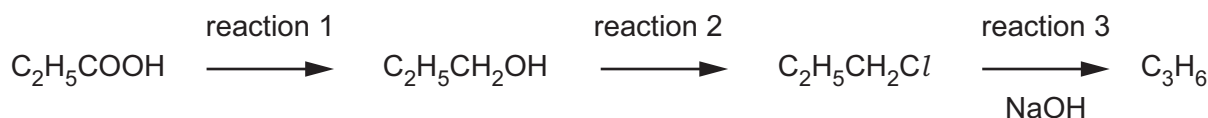
[1]

(ii) Construct an equation to represent reaction 3. Use [O] to represent an atom of oxygen from the oxidising agent.

[1]

8. O/N 25/P23/Q4/a

also Unit 15 (2 marks), Unit 18 (1 mark); this unit 2 marks

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .**Fig. 4.1****(a) (i)** Identify the type of reaction that occurs in reaction 1.

[1]

(ii) Suggest a suitable reagent for reaction 2.

[1]

(iii) Reaction 3 is an elimination reaction.

Write an equation for this reaction and identify the solvent and conditions used.

equation

solvent and conditions

[2]

(iv) $C_2H_5CH_2OH$ can be directly converted to C_3H_6 .

Suggest the reagent and conditions for this conversion.

[1]

9. O/N 25/P24/Q2/c(ii)**2** Aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , are both used as reagents and catalysts.**(b)** A 5.00 dm^3 sealed flask contains 0.400 mol of $CO(g)$ and 0.800 mol of $H_2(g)$ and an Al_2O_3 catalyst. The flask is heated to a temperature of 290°C and allowed to reach equilibrium. Equation 1 shows the reaction.The equilibrium constant, K_c , of equation 1 is given.

$$K_c = \frac{[CH_3OH]}{[CO][H_2]^2}$$

(c) P_4O_{10} catalyses the reversible reaction of CO with H_2 to form CH_3OH . P_4O_{10} then acts as a dehydrating agent, causing CH_3OH to form CH_3OCH_3 .**(ii)** Construct an equation for the dehydration reaction of CH_3OH to form CH_3OCH_3 .

[1]

10. O/N 25/P24/Q3/a(i,ii,iii),b(i)

3 Propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$, is sometimes added to fuel to help it burn.

Fig. 3.1 shows some reactions of propan-2-ol.

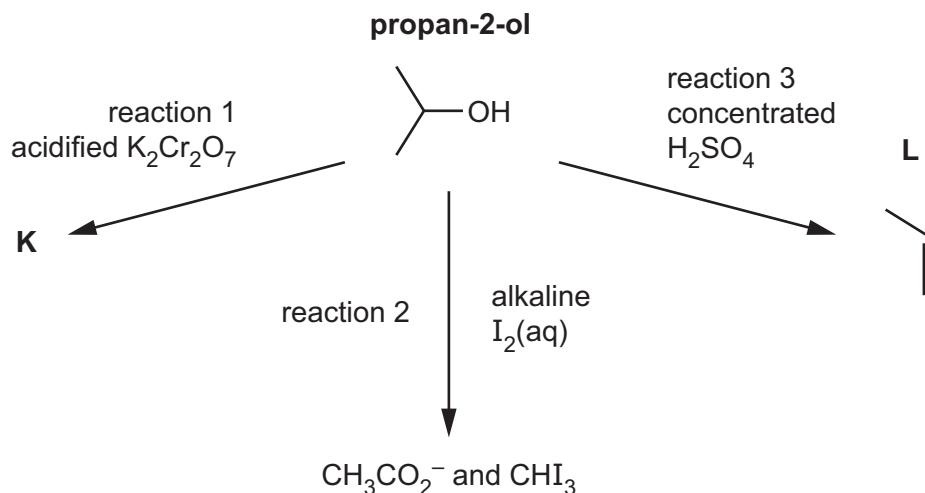


Fig. 3.1

(a) (i) Draw the structure of organic compound **K**.

[1]

(ii) State an observation you would make in reaction 2.

[1]

(iii) State the type of reaction that is shown in reaction 3.

[1]

(b) Propan-2-ol reacts with sodium to produce $(\text{CH}_3)_2\text{CH-O}^-$ anions.

These anions react with 2-bromopropane to form compound **N**, as shown in Fig. 3.3.

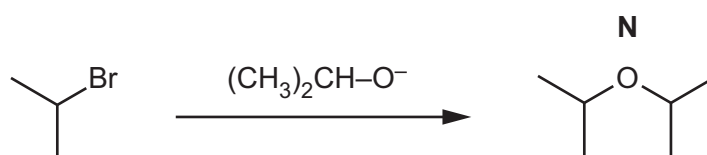


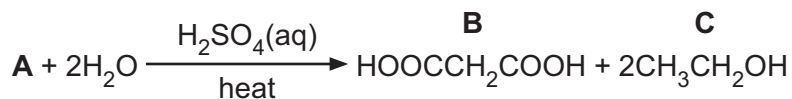
Fig. 3.3

(i) Write an equation for the reaction of propan-2-ol with sodium.

[1]

11. M/J 25/P21/Q5/c(ii),d(ii),e(iii)

- 5** Compound **A** contains the elements carbon, hydrogen and oxygen only.
When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

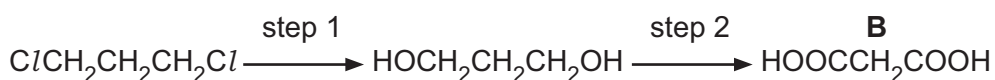
**Fig. 5.1**

- (c) When aqueous Na_2CO_3 is added to separate samples of **B** and **C**, effervescence is observed with **B** only.

(ii) Suggest why **C** does **not** react with aqueous Na_2CO_3 in a similar type of reaction to **B**.

[1]

- (d) A student suggests a two-step synthesis to produce **B**, as shown in Fig. 5.2.

**Fig. 5.2**

- (ii) Identify the type of reaction that $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{OH}$ undergoes in step 2.

[1]

- (e) **B** reacts with an excess of reducing agent **R** to produce **D**.

An excess of PCl_5 is added to **D**.

- A vigorous reaction occurs.
- Misty fumes are seen.
- Organic compound **E** is produced.

- (iii) Name organic compound **E**.

[1]

12. M/J 25/P21/Q6/c(iii)

- 6** Three bottles of colourless liquids labelled **F**, **G** and **H** contain separate pure samples of the compounds ethanal, propanal or propanone but **not** necessarily in that order.

- (c) Separate samples of **F**, **G** and **H** are each tested with Tollens' reagent and with alkaline $I_2(aq)$. The observations are shown in Table 6.1.

Table 6.1

	Tollens' reagent	alkaline $I_2(aq)$
F	no observable change	pale yellow precipitate
G	silver mirror	pale yellow precipitate
H	silver mirror	no precipitate

- (iii) Compound **J** does **not** contain the same functional group as **F**, **G** and **H**. Compound **J** also reacts with alkaline $I_2(aq)$ to produce a pale yellow precipitate.

Suggest the structure of compound **J**.

[1]

13. M/J 25/P22/Q4/b(i),c(ii,iii),d

(b) Propan-1-ol is converted to compound **T** in a three-step synthesis.

In step 1, SOCl_2 is added to propan-1-ol to produce compound **E**.

In step 2, **E** reacts with a suitable reagent to produce butanenitrile.

In step 3, butanenitrile is heated with NaOH(aq) .

(i) Name the type of reaction that occurs in step 1.

[1]

(c) Compounds **A**, **B** and **C** belong to the alcohol homologous series. Each molecule of **A**, **B** and **C** contains four saturated carbon atoms.

(ii) Identify the gas produced when Na(s) is added to separate samples of each alcohol.

[1]

(iii) Describe the role of Na(s) when it reacts with alcohols.

[1]

(d) Table 4.1 shows the results of two tests on separate samples of **A**, **B** and **C**.

Table 4.1

compound	heat under reflux with acidified $\text{K}_2\text{Cr}_2\text{O}_7$	warm with alkaline $\text{I}_2(\text{aq})$
A	remains orange	no visible change
B	orange to green	no visible change
C	orange to green	pale yellow precipitate

(i) Deduce the structure of **A**.

[1]

(ii) Deduce a possible name for **B**.

[1]

(iii) Deduce the identity of **C**.

[1]

14. O/N 24/P22/Q2/b(iv)

2 Oxygen is a Group 16 element.

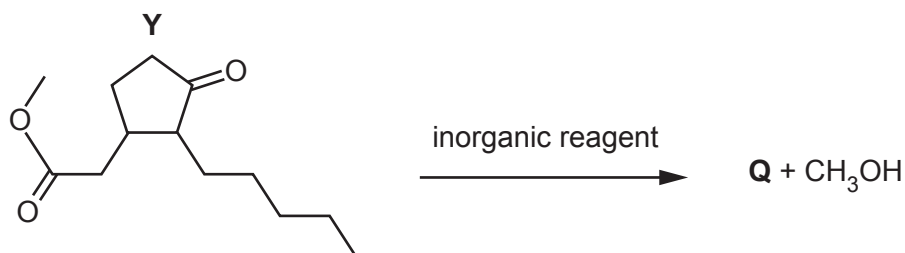
(b) H_2O reacts with both inorganic and organic compounds.

(iv) Explain why alcohols are less acidic than water.

[2]

15. M/J 24/P21/Q6/c(iii)

- (c) Both functional groups in one molecule of **Y** react with an inorganic reagent to form one molecule of **Q** and one molecule of methanol, CH_3OH , as shown in Fig. 6.3.

**Fig. 6.3**

- (iii) Under certain conditions, 0.0020 mol of **Q** reacts with an excess of sodium to produce a total of 44.8 cm³ of gas at s.t.p.

Calculate the number of hydroxyl groups present in a molecule of **Q**.

Show your working.

number of hydroxyl groups = [2]

16. O/N 23/P21/Q3/e(i,ii)

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(e) The structure of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is shown in Fig. 3.2.

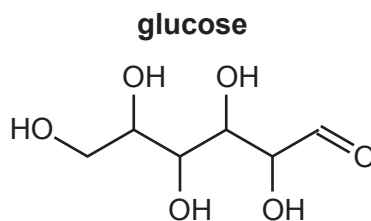


Fig. 3.2

- (i)** Complete Table 3.1 to identify the number of primary, secondary and tertiary alcohol groups present in the structure shown in Fig. 3.2.

Table 3.1

type of alcohol group	primary	secondary	tertiary
number of groups			

[1]

- (ii)** Separate samples of aqueous glucose are tested with the reagents shown in Table 3.2.

Complete Table 3.2 with the observation for each reaction.

Write “no reaction” if applicable.

Table 3.2

reagent and conditions	observation with glucose
acidified $\text{KMnO}_4(\text{aq})$ and warm	
Fehling's reagent and warm	
alkaline $\text{I}_2(\text{aq})$ and warm	

[3]

17. O/N 23/P22/Q3/c(ii)

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

(c) Fig. 3.1 shows a reaction scheme that involves H_3PO_4 in several reactions.

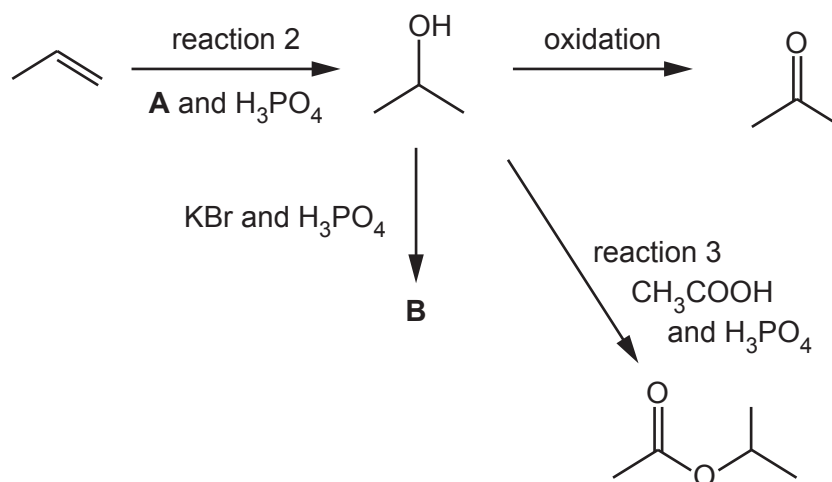


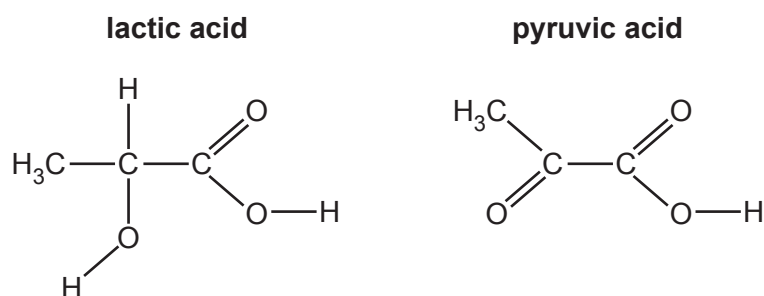
Fig. 3.1

(ii) Draw the structure of **B**.

[1]

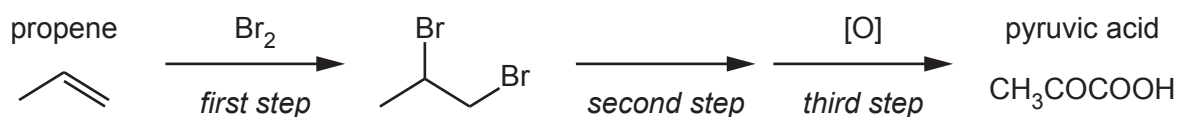
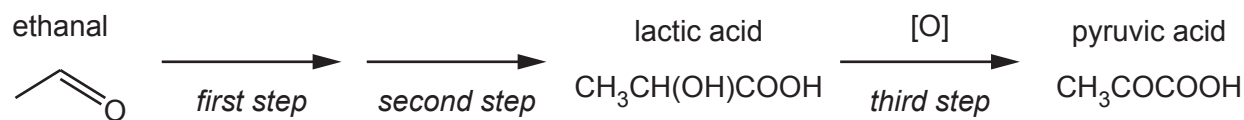
18. O/N 23/P22/Q4/b(ii)

- 4 Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, and pyruvic acid, CH_3COCOOH , both contain two functional groups.

**Fig. 4.1**

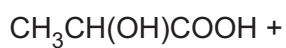
- (b) Two possible syntheses of pyruvic acid are shown in Fig. 4.3 and Fig. 4.4.

Each synthesis has a total of three steps.

**Fig. 4.3****Fig. 4.4**

- (ii) Write an equation for the oxidation of lactic acid to pyruvic acid, the third step of Fig. 4.4.

Use [O] to represent one atom of oxygen from an oxidising agent.



[1]

19. O/N 23/P23/Q3/e(i,ii)

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(e) The structure of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is shown in Fig. 3.2.

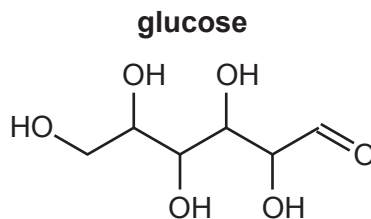


Fig. 3.2

- (i)** Complete Table 3.1 to identify the number of primary, secondary and tertiary alcohol groups present in the structure shown in Fig. 3.2.

Table 3.1

type of alcohol group	primary	secondary	tertiary
number of groups			

[1]

- (ii)** Separate samples of aqueous glucose are tested with the reagents shown in Table 3.2.

Complete Table 3.2 with the observation for each reaction.

Write “no reaction” if applicable.

Table 3.2

reagent and conditions	observation with glucose
acidified $\text{KMnO}_4(\text{aq})$ and warm	
Fehling's reagent and warm	
alkaline $\text{I}_2(\text{aq})$ and warm	

[3]

20. M/J 23/P21/Q4/b(i),c

4 **V** is a colourless liquid.

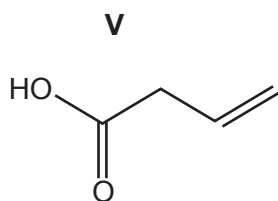


Fig. 4.1

(b) **V** reacts to form **Z** in a single reaction, as shown in Fig. 4.2.

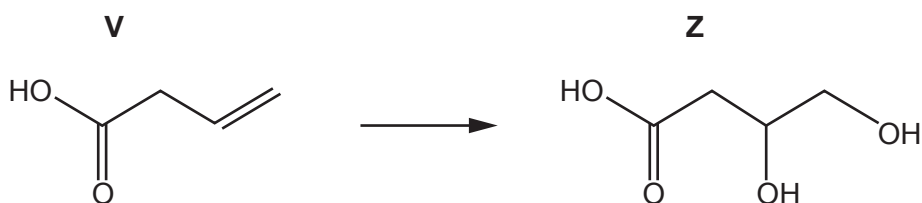


Fig. 4.2

(i) Suggest the reagent and conditions needed to form **Z** from **V**.

[1]

(c) **Q** contains the elements carbon, hydrogen and oxygen only. It is a saturated molecule with no branching in its carbon backbone.

Q contains only one functional group.

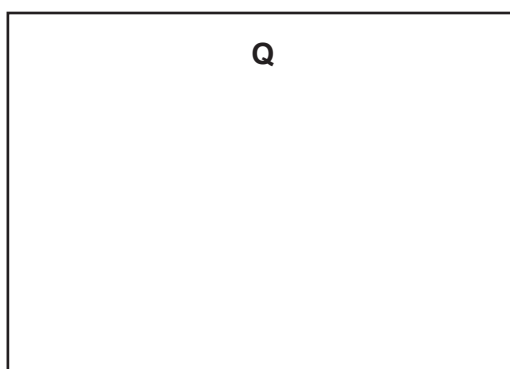
The relative molecular mass of **Q** is 88.

No effervescence is seen when Na_2CO_3 is added to **Q**.

Effervescence is seen when sodium is added to **Q**.

Q reacts with alkaline $\text{I}_2(\text{aq})$ to form a yellow precipitate.

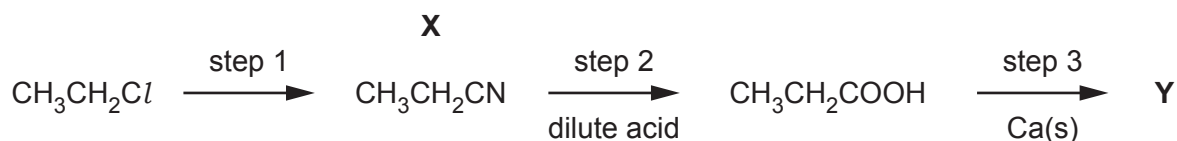
Draw the structure of **Q** in the box.



[2]

21. M/J 23/P22/Q3/b

- 3 Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

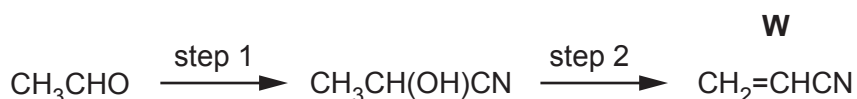
- (b) $\text{CH}_3\text{CH}_2\text{COOH}$ can also be formed from propan-1-ol and potassium dichromate(VI). State the conditions required.

[1]

22. M/J 23/P23/Q6/b(iii),f(i)

- 6 Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

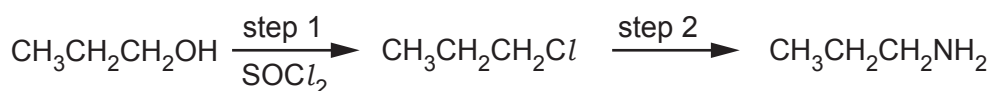
- (b) CH_3CHO is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1. In step 1, CH_3CHO is heated with HCN in the presence of KCN .

**Fig. 6.1**

- (iii) Suggest a suitable reagent and conditions for step 2 in Fig. 6.1.

[1]

- (f) Propylamine can also be formed in a two-step synthesis from propan-1-ol, as shown in Fig. 6.4.

**Fig. 6.4**

- (i) Name the type of reaction in step 1 in Fig. 6.4.

[1]

23. MAR 23/P22/Q4/a(iii),b(ii)

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

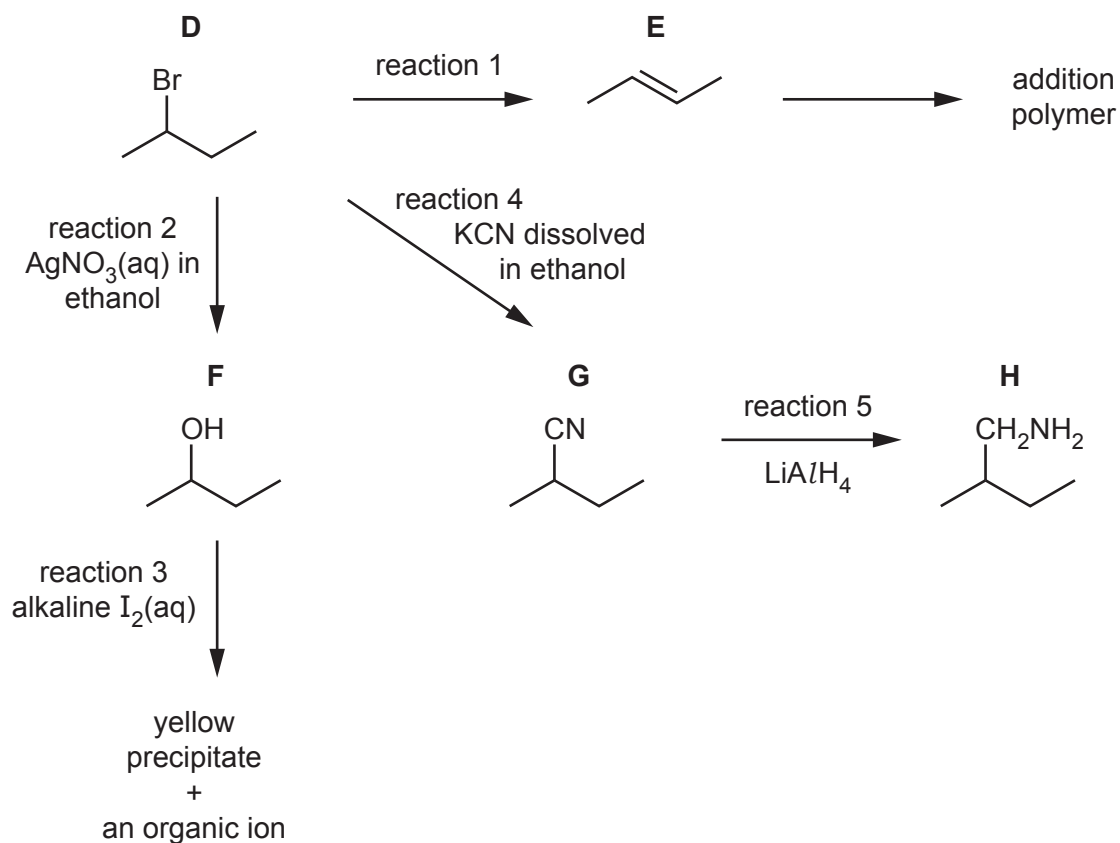


Fig. 4.1

(iii) **E** also forms when **F** is heated strongly in the presence of an Al_2O_3 catalyst.

Write an equation for this reaction.

[1]

(ii) Identify the yellow precipitate and the organic ion formed in reaction 3.

yellow precipitate

organic ion

[2]

24. O/N 22/P21/Q3/c

- 3** Some of the common chlorides of Period 3 elements are shown in the list.



(c) PCl_5 reacts with alcohols to form chloroalkanes.

- (i)** Identify this type of reaction.

[1]

- (ii)** Draw the structure of the organic product formed in the reaction of an excess of PCl_5 with butane-1,3-diol.

[1]

25. O/N 22/P23/Q3/c

- 3** Some of the common chlorides of Period 3 elements are shown in the list.



(c) PCl_5 reacts with alcohols to form chloroalkanes.

- (i)** Identify this type of reaction.

[1]

- (ii)** Draw the structure of the organic product formed in the reaction of an excess of PCl_5 with butane-1,3-diol.

[1]

26. M/J 22/P21/Q3/d(i,ii)

- 3** Liquids that contain molecules of **T** smell like lemons.

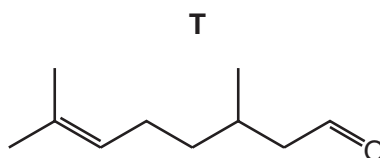


Fig. 3.1

(c) Fig. 3.2 shows two reactions of **T**.

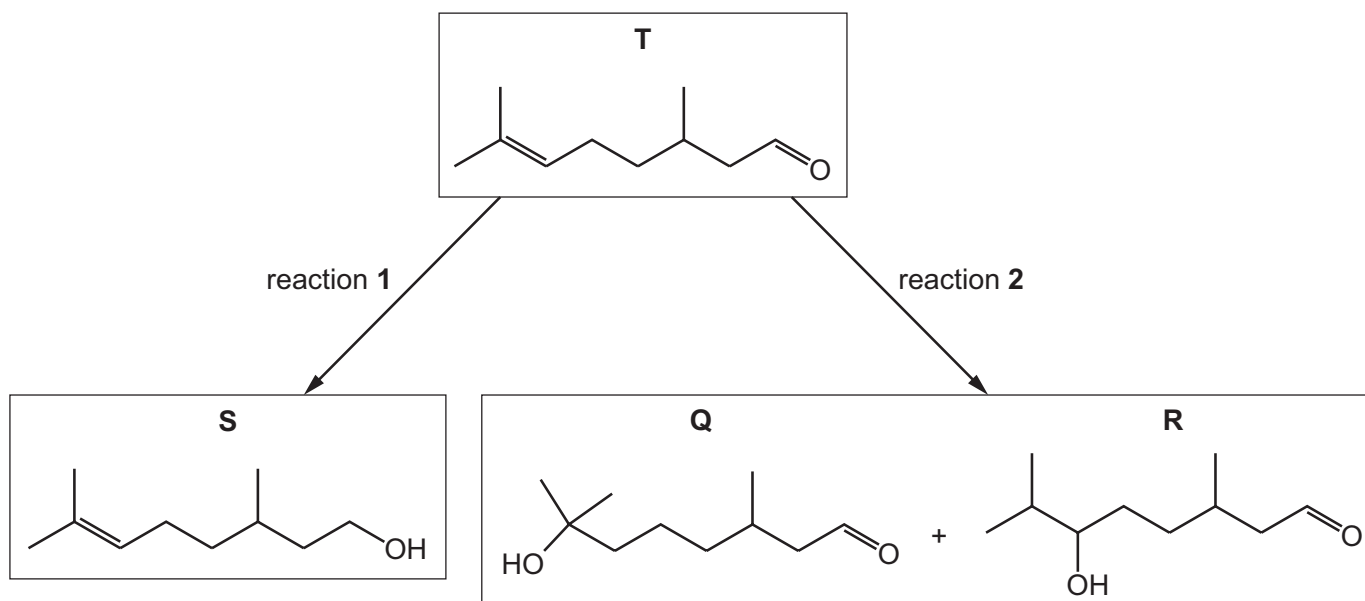


Fig. 3.2

(d) Separate samples of **Q** and **R** are added to separate test-tubes containing acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ and heated.

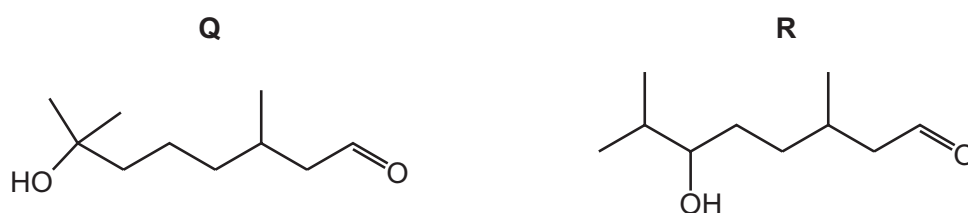


Fig. 3.3

(i) Predict the observations for each test-tube. Explain your answer in terms of the functional groups present in **Q** and **R**.

[3]

(ii) When $\text{PCl}_5(\text{s})$ is added to separate samples of **Q** and **R** at room temperature, both react vigorously.

Complete the equation shown in Fig. 3.4 to describe the reaction that occurs when **R** reacts with $\text{PCl}_5(\text{s})$.

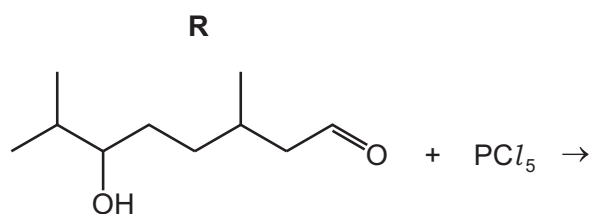


Fig. 3.4

[2]

27. M/J 22/P23/Q4/b(iv)

- (b) Two bottles labelled **Q** and **M** each contain a straight-chain halogenoalkane with molecular formula C_4H_9X , where **X** represents Cl, Br or I.

A sample from each bottle is added to separate samples of equal amounts of aqueous silver nitrate in ethanol. In each reaction, the same organic product, **T**, and a precipitate are made, as shown in Fig. 4.2.

**Fig. 4.2**

Table 4.1 describes the colour of each of the precipitates made.

Table 4.1

halogenoalkane added to $\text{AgNO}_3(\text{aq})$ in ethanol	colour of precipitate
Q	white
M	yellow

- (iv) When pure **T** is added to alkaline $\text{I}_2(\text{aq})$, a yellow precipitate and an anion, **L**, are made.

Identify the anion **L**.

[1]

28. M/J 22/P23/Q5/a(ii)

- 5 (a) But-2-ene reacts with KMnO_4 to form organic product, **Y**.

Y does not react with Na_2CO_3 .

Gas is produced when an excess of Na is added to **Y**.

- (ii) 24.0cm^3 of gas is produced when an excess of Na is added to 0.001mol of **Y**, when measured under room conditions.

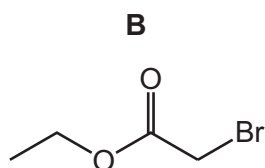
Assume that 1mol of gas occupies 24.0dm^3 under room conditions.

Deduce a possible structure of **Y**. Explain your answer.

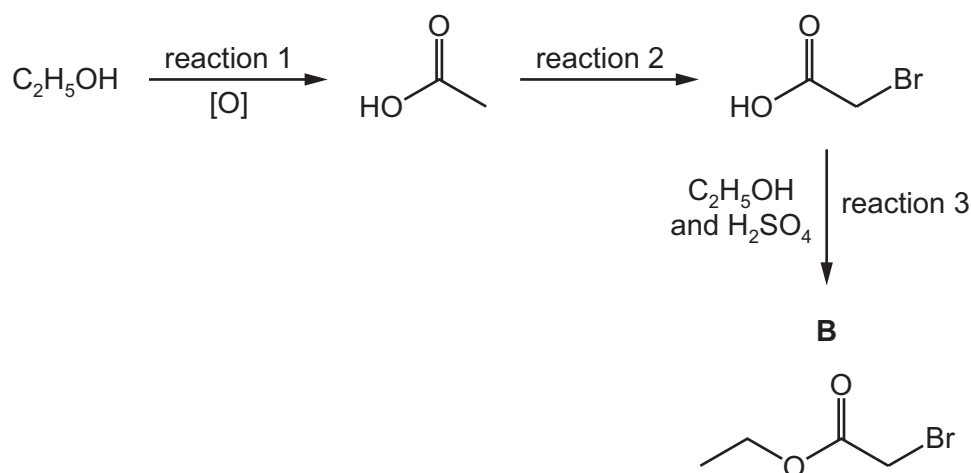
[3]

29. O/N 21/P21/Q4/a(i,ii)

4 Compound **B** is a liquid with a fruity smell.



The reaction scheme shows how **B** can be made from ethanol, C_2H_5OH .



(a) (i) Reaction 1 is an oxidation reaction.

Give the reagent(s) and conditions required for reaction 1.

reagent(s)

conditions

[2]

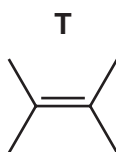
(ii) Construct an equation to represent reaction 1.

Use $[O]$ to represent an oxygen atom from the oxidising agent in this reaction.

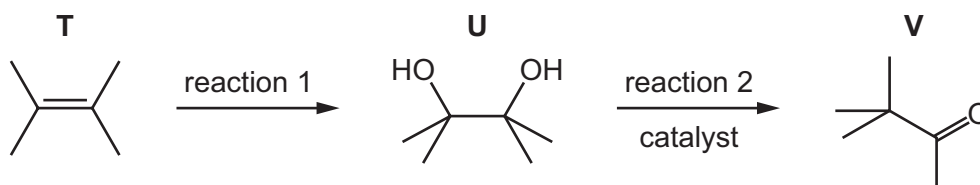
[1]

30. O/N 21/P22/Q3/e(v)

3 Compound **T** is an isomer of C_6H_{12} .

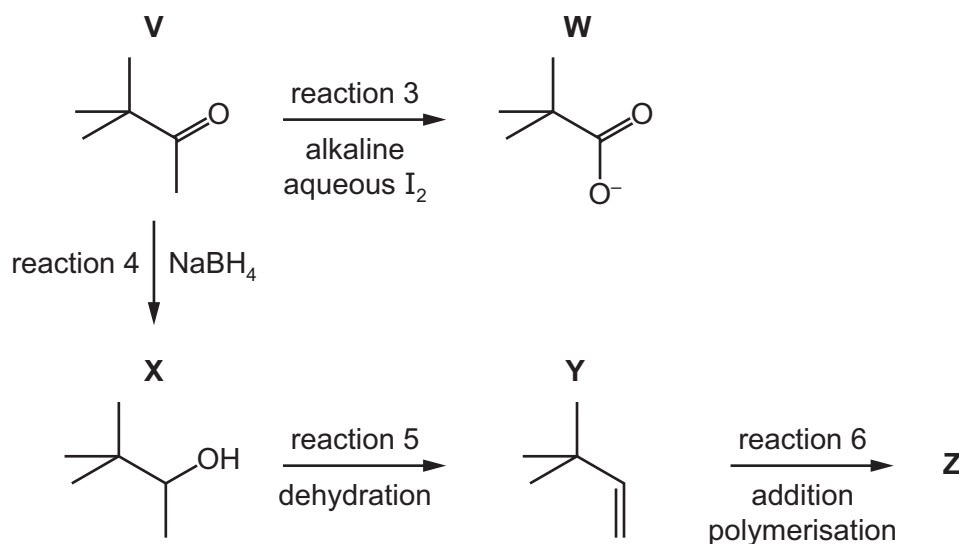


(d) A reaction scheme starting with **T** is shown. Reaction 2 occurs in the presence of a catalyst; knowledge of the mechanism for this reaction is not required.



(e) **V** is used in a wide range of organic reactions.

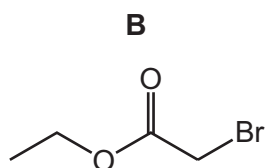
Some reactions of **V** are shown.



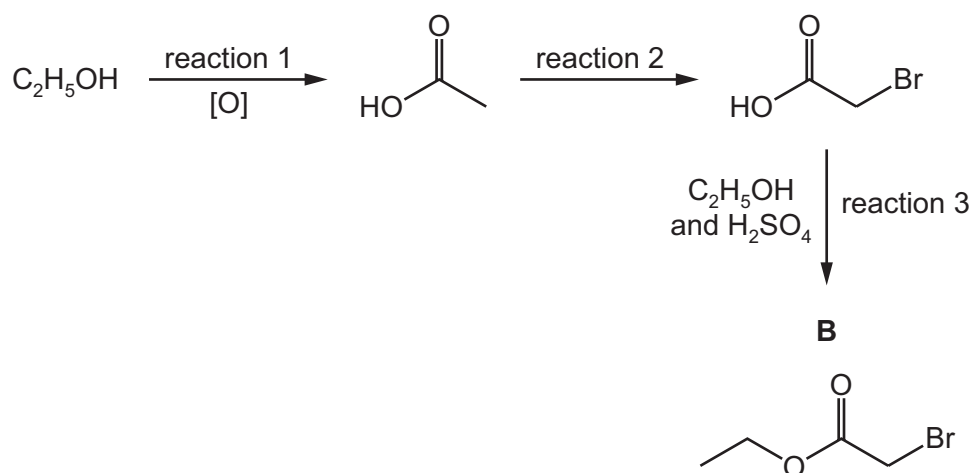
(v) Both optical isomers of **X** can be dehydrated to form a single product, **Y**.
Give the reagent(s) and conditions required for reaction 5.

31. O/N 21/P23/Q4/a(i,ii)

4 Compound **B** is a liquid with a fruity smell.



The reaction scheme shows how **B** can be made from ethanol, C_2H_5OH .



(a) (i) Reaction 1 is an oxidation reaction.

Give the reagent(s) and conditions required for reaction 1.

reagent(s)

conditions

[2]

(ii) Construct an equation to represent reaction 1.

Use [O] to represent an oxygen atom from the oxidising agent in this reaction.

[1]

32. M/J 21/P22/Q4/c,d

(c) A sample of propan-1-ol reacts with concentrated sulfuric acid to form propene.

Identify the role of concentrated sulfuric acid in this reaction.

[1]

(d) Alcohol **Y** reacts completely when warmed with acidified $\text{Cr}_2\text{O}_7^{2-}$ to form **Z**.

Z is distilled from the reaction mixture as soon as it is made.

Tollens' reagent is added to a sample of **Z** and warmed. A silver mirror forms.

(i) Name the type of reaction that occurs when **Y** reacts to form **Z**.

[1]

(ii) Identify with a tick (✓) the functional group(s) present in **Z**.

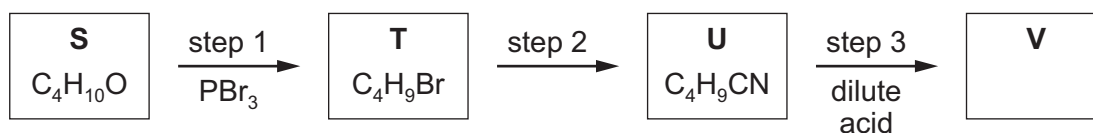
functional group	present in Z
aldehyde	
ketone	
carboxylic acid	

[1]

33. M/J 21/P22/Q5/b(ii)

5 **S** is a secondary alcohol with molecular formula $\text{C}_4\text{H}_{10}\text{O}$.

(b) **S** is converted to **V** in a three-step reaction sequence.



In step 1, the secondary alcohol **S** reacts with PBr_3 to produce **T**, which has molecular formula $\text{C}_4\text{H}_9\text{Br}$.

(ii) Name the type of reaction that occurs in step 1.

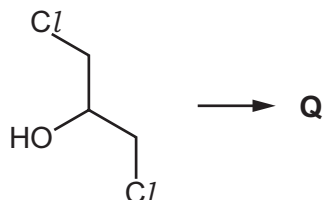
[1]

34. M/J 21/P23/Q4/b

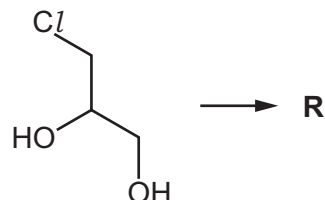
- (b) Separate samples of 1,3-dichloropropan-2-ol and 3-chloropropane-1,2-diol are heated with excess acidified $\text{Cr}_2\text{O}_7^{2-}$ until there is no further reaction.

In each reaction, a different organic product, **Q** or **R**, is made.

1,3-dichloropropan-2-ol



3-chloropropane-1,2-diol



Q and **R** are tested separately with 2,4-dinitrophenylhydrazine solution, 2,4-DNPH, and sodium carbonate solution, $\text{Na}_2\text{CO}_3(\text{aq})$.

Complete the table to give any relevant observations.

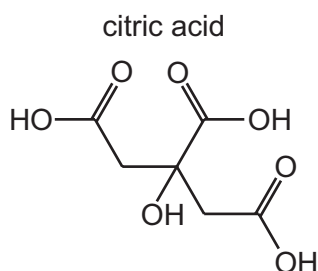
If no reaction occurs, write 'no visible change'.

reagent	observation with Q	observation with R
2,4-DNPH		
$\text{Na}_2\text{CO}_3(\text{aq})$		

[4]

35. M/J 21/P23/Q5/b(ii)

- 5 Compound **X** contains the same functional groups as citric acid.



The table describes some of the similarities and differences between citric acid and compound **X**.

	citric acid	X
chiral centre	no	yes
reaction with Na	fizzing	fizzing
reaction with $\text{H}^+/\text{Cr}_2\text{O}_7^{2-}$	remains orange	orange to green

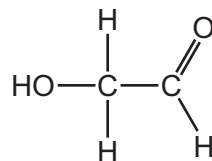
- (ii) Explain why compound **X** reacts with acidified $\text{Cr}_2\text{O}_7^{2-}$ but citric acid does not.

[2]

36. MAR 21/P22/Q4/c

- 4 Hydroxyethanal, HOCH_2CHO , has been observed in dust clouds near the centre of our galaxy.

hydroxyethanal



- (c) Hydroxyethanal is converted to ethanedioic acid, $(\text{CO}_2\text{H})_2$, when it reacts with excess acidified dichromate(VI) ions, $\text{Cr}_2\text{O}_7^{2-}$.

- (i) State the role of acidified $\text{Cr}_2\text{O}_7^{2-}$ in this reaction.

[1]

- (ii) State and explain any other necessary conditions for this reaction to be successful.

[2]

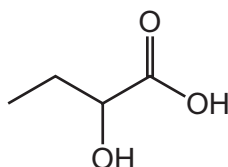
37. O/N 20/P21/Q3/c

also Unit 17 (4 marks), Unit 22 (2 marks), Unit 19 (1 mark); this unit 4 marks

- 3 The reducing agent LiAlH_4 can be synthesised by reacting aluminium chloride with lithium hydride, LiH .

- (c) Two students try to prepare 2-hydroxybutanoic acid in the laboratory.

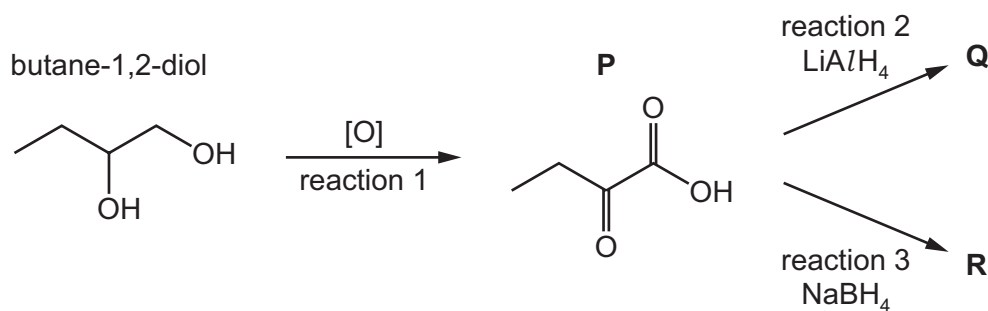
2-hydroxybutanoic acid



Both students oxidise butane-1,2-diol to form **P** in reaction 1.

One student then reduces **P** using LiAlH_4 . **Q** is formed.

The other student reduces **P** using NaBH_4 . **R** is formed.



- (i) State the reagents and conditions required for reaction 1.

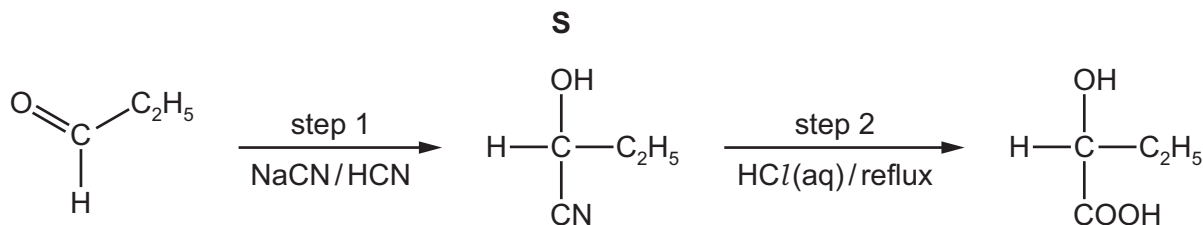
[2]

- (ii) Only one of the students successfully prepares 2-hydroxybutanoic acid.

Identify which of **Q** or **R** is 2-hydroxybutanoic acid and explain the difference between reactions 2 and 3.

[2]

A third student prepares 2-hydroxybutanoic acid using propanal as the starting material. In step 1 the student reacts propanal with a mixture of NaCN and HCN.



(iii) Draw the mechanism for the reaction of propanal with the mixture of NaCN and HCN to form **S**.

- Identify the ion that reacts with propanal.
- Draw the structure of the intermediate of the reaction.
- Include all charges, partial charges, lone pairs and curly arrows.



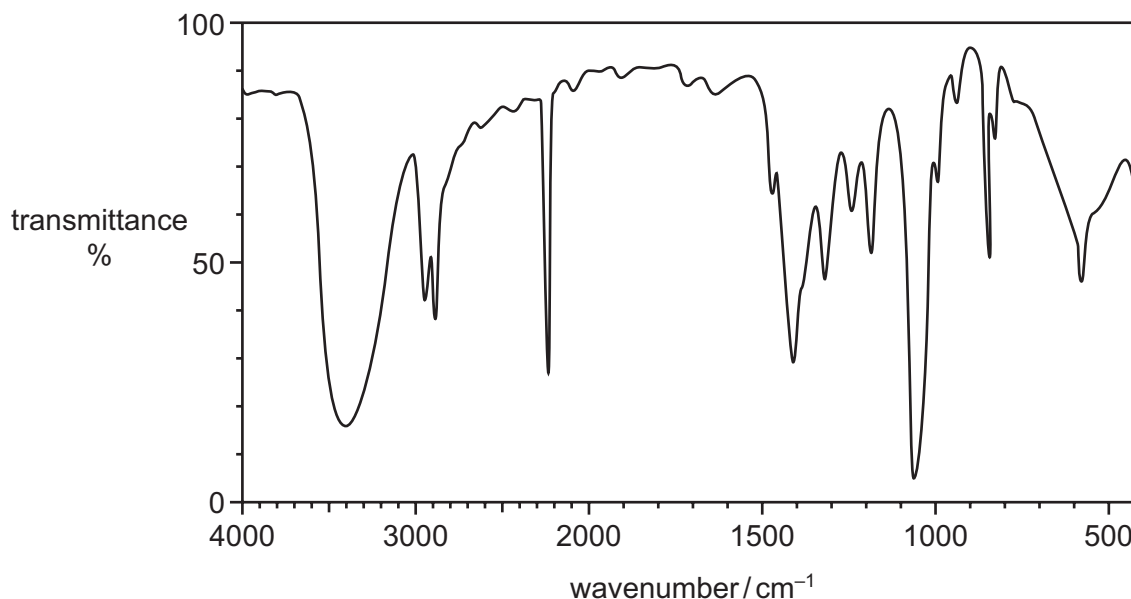
[4]

(iv) Complete the equation for the reaction in step 2, when **S** is heated under reflux with HCl(aq).



[1]

(v) The infrared spectrum of an organic compound is shown. The organic compound is either **S** or 2-hydroxybutanoic acid.



Deduce the identity of the compound. Give **two** reasons for your answer.

In your answer, identify any relevant absorptions **above 1500 cm^{-1}** in the spectrum and the bonds that correspond to these absorptions.

[2]

38. O/N 20/P21/Q4/b(i)

4 Iodine is used in many inorganic and organic reactions.

(b) Iodoalkanes contain carbon-iodine bonds.

The simplest iodoalkane is CH_3I .

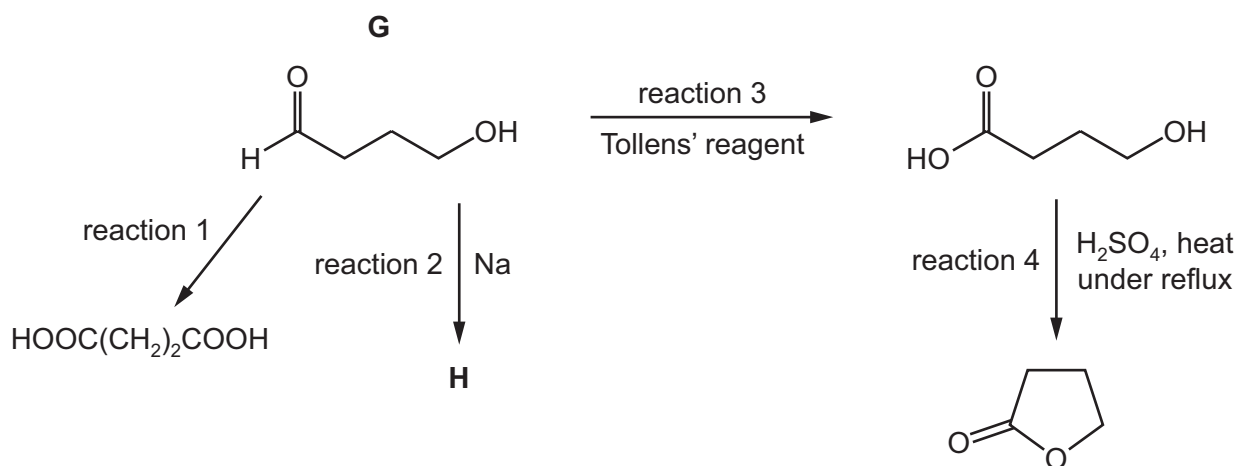
(i) CH_3I can be made from methanol, CH_3OH .

Identify a reagent that can convert CH_3OH to CH_3I .

[1]

39. O/N 20/P22/Q4/a(i,ii,iii,v)

4 Some reactions of compound **G** are shown.



(a) (i) State the type of reaction that occurs in reaction 1.

[1]

(ii) Suggest the reagent(s) and conditions required for reaction 1.

[2]

(iii) Draw the structure of the organic product, **H**, from reaction 2.

[1]

(v) Give the type of reaction shown by reaction 4.

[1]

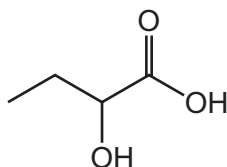
40. O/N 20/P23/Q3/c

also Unit 17 (4 marks), Unit 22 (2 marks), Unit 19 (1 mark); this unit 4 marks

- 3** The reducing agent LiAlH_4 can be synthesised by reacting aluminium chloride with lithium hydride, LiH .

(c) Two students try to prepare 2-hydroxybutanoic acid in the laboratory.

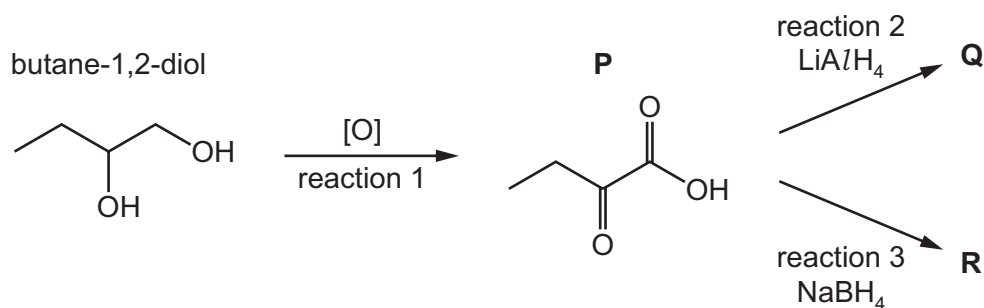
2-hydroxybutanoic acid



Both students oxidise butane-1,2-diol to form **P** in reaction 1.

One student then reduces **P** using LiAlH_4 . **Q** is formed.

The other student reduces **P** using NaBH_4 . **R** is formed.



- (i) State the reagents and conditions required for reaction 1.

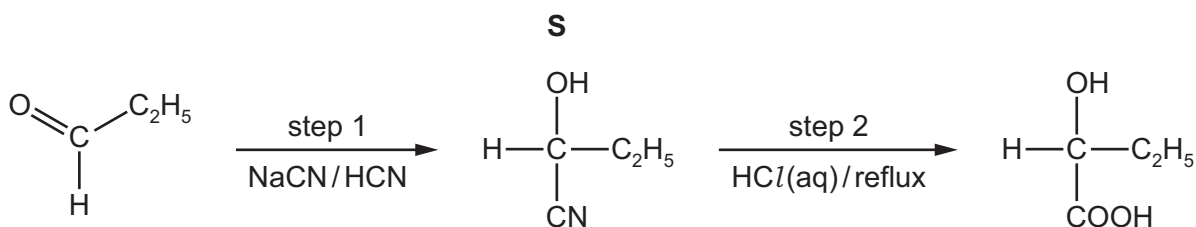
[2]

- (ii) Only one of the students successfully prepares 2-hydroxybutanoic acid.

Identify which of **Q** or **R** is 2-hydroxybutanoic acid and explain the difference between reactions 2 and 3.

[2]

A third student prepares 2-hydroxybutanoic acid using propanal as the starting material. In step 1 the student reacts propanal with a mixture of NaCN and HCN .



- (iii) Draw the mechanism for the reaction of propanal with the mixture of NaCN and HCN to form **S**.

- Identify the ion that reacts with propanal.
- Draw the structure of the intermediate of the reaction.
- Include all charges, partial charges, lone pairs and curly arrows.



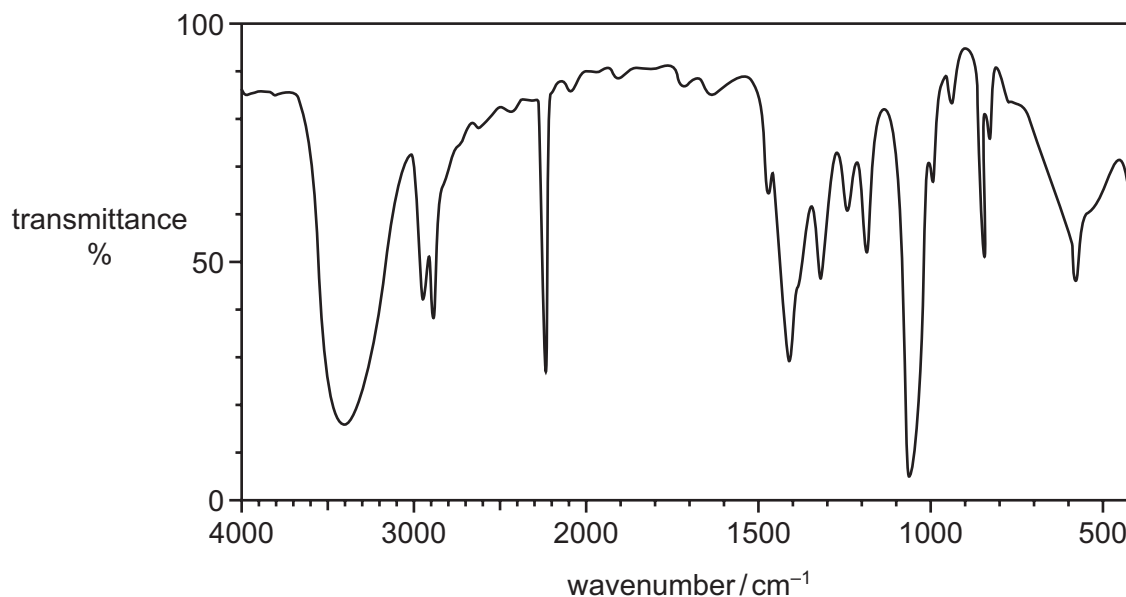
[4]

- (iv) Complete the equation for the reaction in step 2, when **S** is heated under reflux with HCl(aq) .



[1]

- (v) The infrared spectrum of an organic compound is shown. The organic compound is either **S** or 2-hydroxybutanoic acid.



Deduce the identity of the compound. Give **two** reasons for your answer.

In your answer, identify any relevant absorptions **above 1500 cm⁻¹** in the spectrum and the bonds that correspond to these absorptions.

[2]

41. O/N 20/P23/Q4/b(i)

4 Iodine is used in many inorganic and organic reactions.

(b) Iodoalkanes contain carbon-iodine bonds.

The simplest iodoalkane is CHI_3 .

(i) CH_3I can be made from methanol, CH_3OH .

Identify a reagent that can convert CH_3OH to CH_3I .

[1]

42. MAR 20/P22/Q2/e(ii,iii)

2 The Group 17 elements, chlorine, bromine and iodine, are non-metals that show trends in their physical and chemical properties.

(e) Lucas's reagent is a mixture of HCl and ZnCl_2 . Primary, secondary and tertiary alcohols can be distinguished by their reaction with Lucas's reagent.

Alcohols react with the HCl in Lucas's reagent to form halogenoalkanes.

ZnCl_2 acts as a homogeneous catalyst for these reactions.

(ii) Pentan-3-ol, $\text{C}_2\text{H}_5\text{CH}(\text{OH})\text{C}_2\text{H}_5$, reacts slowly with HCl to form a secondary halogenoalkane.

Complete the equation for this reaction using structural formulae.

$\text{C}_2\text{H}_5\text{CH}(\text{OH})\text{C}_2\text{H}_5 + \dots\dots\dots$ [1]

(iii) The fastest reaction shown by Lucas's reagent is with a tertiary alcohol.

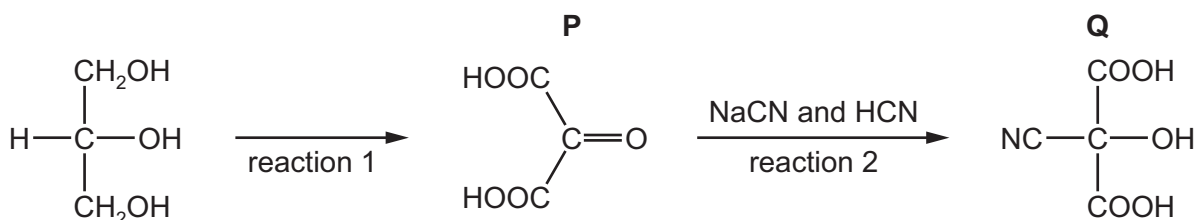
Draw the structure of the tertiary alcohol that is an isomer of pentan-3-ol.

[1]

43. MAR 20/P22/Q3/a(i)

3 Glycerol, $\text{CH}_2(\text{OH})\text{CH}(\text{OH})\text{CH}_2\text{OH}$, is widely used in the food industry and in pharmaceuticals.

(a) A series of reactions starting from glycerol is shown.

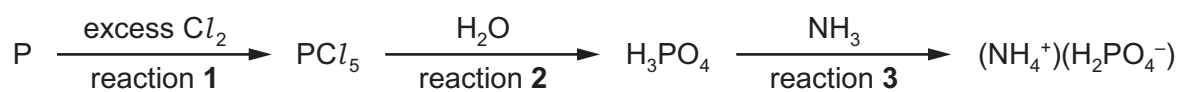


(i) Suggest the reagent(s) and conditions for reaction 1.

[2]

44. O/N 19/P22/Q3/c(i,ii)

3 A series of reactions for phosphorus and its compounds is shown.



(c) PCl_5 can be used to convert alcohols to halogenoalkanes.

(i) Write an equation for the reaction of $\text{C}_2\text{H}_5\text{OH}$ with PCl_5 to form $\text{C}_2\text{H}_5\text{Cl}$.

[1]

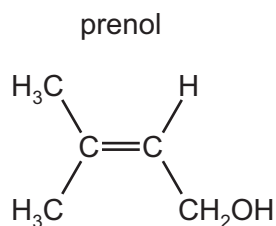
(ii) State the type of reaction in **(i)**.

[1]

45. O/N 19/P22/Q4/b

also Unit 14 (1 mark), Unit 13 (1 mark); this unit 2 marks

- 4** Prenol is a naturally occurring organic molecule found in many fruits. It contains both an alkene and an alcohol functional group.



- (b)** Prenol reacts with steam to form a mixture of three isomers, **J**, **K** and **L**, of molecular formula $C_5H_{12}O_2$.

- (i)** When **J** is heated with excess acidified potassium dichromate(VI) it forms an organic product which shows no reaction with 2,4-DNPH.

Draw the structure of **J**.

[1]

K and **L** are stereoisomers with molecular formula $C_5H_{12}O_2$.

K and **L** both react when heated with excess acidified potassium dichromate(VI) to form **M**, $C_5H_8O_3$.

M forms an orange precipitate on reaction with 2,4-DNPH.

- (ii)** Give the structural formula of **K** and **L**.

[1]

- (iii)** Name the type of stereoisomerism shown by **K** and **L**.

[1]

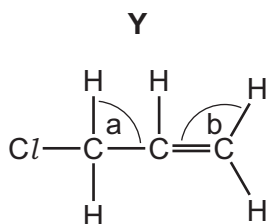
- (iv)** Give the balanced equation to represent the reaction of **K**, $C_5H_{12}O_2$, with acidified potassium dichromate(VI) to form **M**, $C_5H_8O_3$.

Use [O] to represent an atom of oxygen provided by the oxidising agent.

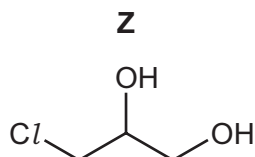
[1]

46. M/J 19/P21/Q4/c(iii)

4 The structure of compound **Y** is shown.



(c) When **Y** reacts with cold, dilute, acidified manganate(VII) ions, compound **Z** is produced.



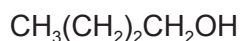
(iii) Alcohols can be classified as primary, secondary or tertiary.

Identify with a tick (✓) the alcohol group(s) present in **Z**.

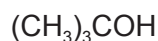
	alcohol group present in Z
primary	
secondary	
tertiary	

47. M/J 19/P23/Q6/b

(b) The structures of **V** and **W** are shown.



V



W

(i) Name the class of compound that **V** and **W** each belong to.

V

W

[2]

(ii) **V** and **W** both react with sodium metal.

Write an equation for the reaction of **V** with sodium metal.

[1]

(iii) Name a reagent used to distinguish **V** from **W**. Describe any observations.

reagent

observations with **V**

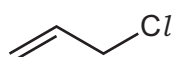
observations with **W**

[3]

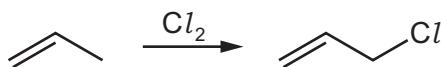
48. MAR 19/P22/Q4/b(iv)

4 Allyl chloride is an important chemical used in the manufacture of plastics, pharmaceuticals and pesticides.

allyl chloride



(b) Allyl chloride can be produced by many different methods. The most common method is chlorination of propene which proceeds via a free-radical substitution mechanism.



(iv) Allyl chloride can also be formed by the following substitution reaction.

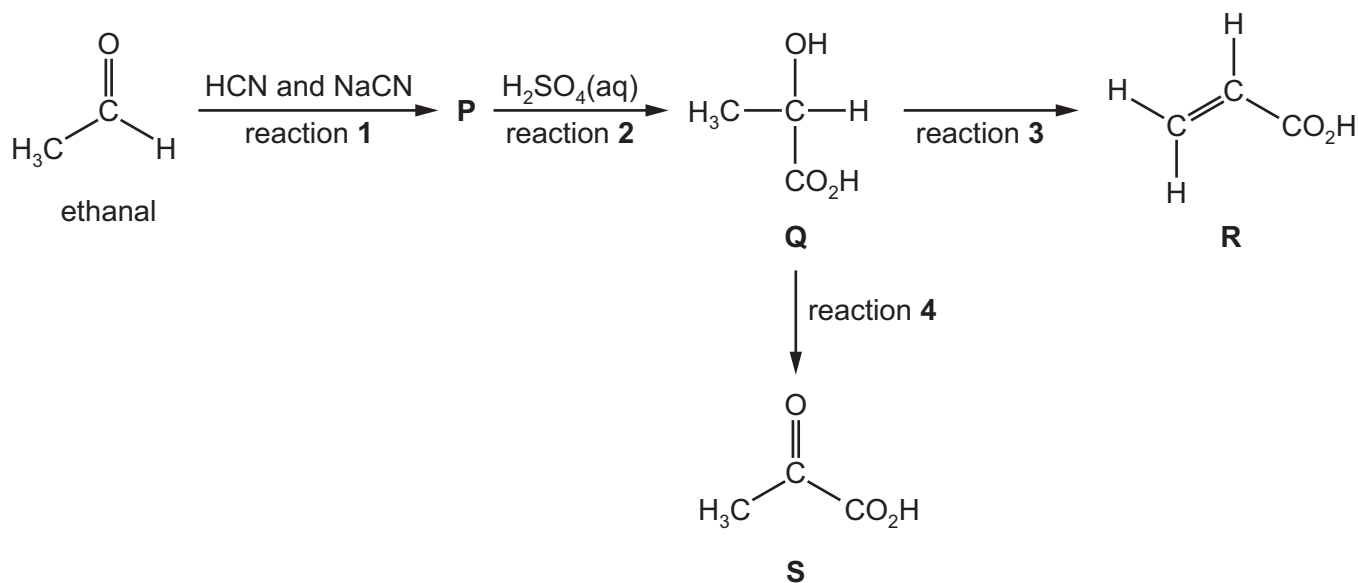


Suggest the identity of reagent **X**.

[1]

49. O/N 18/P22/Q4/a(iii,iv)

4 The diagram shows a reaction sequence starting from ethanal.



(iii) Write an equation to represent reaction 4.

Use [O] to represent the oxidising agent.

[1]

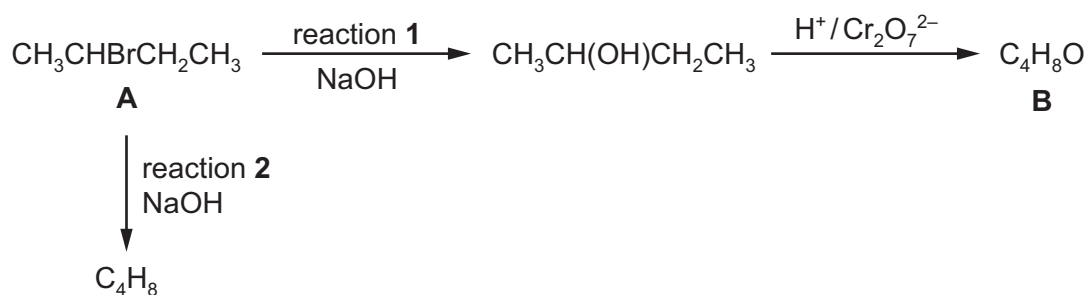
(iv) State the reagents and conditions for reaction 4.

[1]

50. M/J 18/P23/Q4/a(ii)

4 A is $\text{CH}_3\text{CHBrCH}_2\text{CH}_3$.

(a) Some reactions of A are shown.

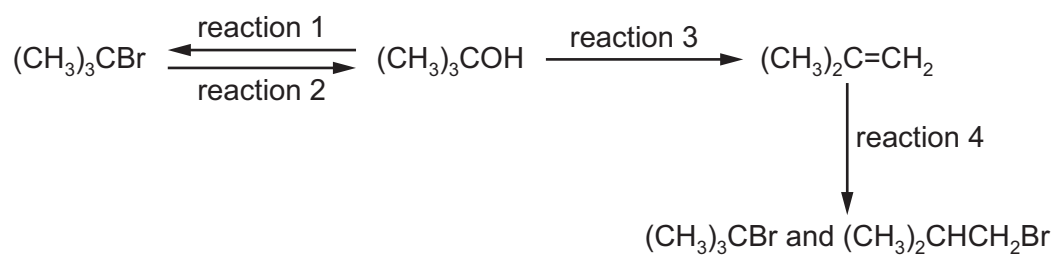


(ii) Name the class of compound to which B belongs.

[1]

51. O/N 17/P22/Q4/a

4 Some reactions are shown, based on methylpropan-2-ol, $(\text{CH}_3)_3\text{COH}$.



- (a)** For each of the reactions state the reagent(s), the particular conditions required, if any, and the type of reaction.

For the type of reaction choose from the list.

Each type may be used once, more than once or not at all.

Each reaction may be described by one or more than one type.

hydrolysis dehydration substitution
 oxidation addition condensation

reaction	reagent(s) and conditions	type(s) of reaction
1		
2		
3		
4		

[5]

52. M/J 17/P22/Q4/b(i)

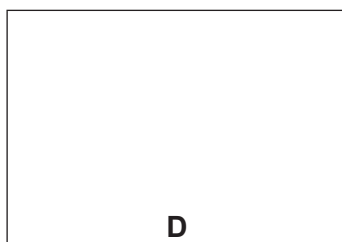
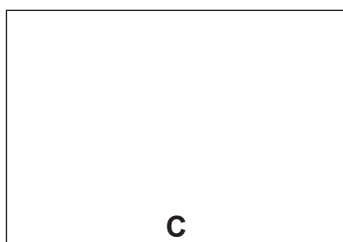
- (b) The alcohols **C** and **D** are isomers of each other with molecular formula $C_4H_{10}O$. Both isomers are branched.

When **C** is heated under reflux with acidified potassium dichromate(VI) no colour change is observed.

When **D** is heated under reflux with acidified potassium dichromate(VI) the colour of the mixture changes from orange to green and **E**, $C_4H_8O_2$, is produced.

E reacts with aqueous sodium carbonate to form carbon dioxide gas.

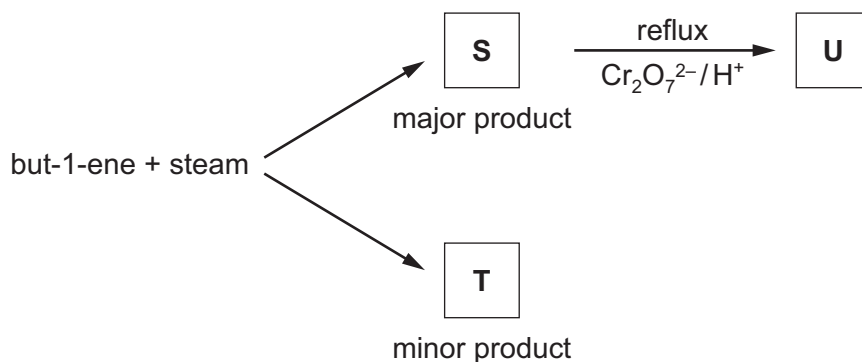
- (i) Identify **C**, **D** and **E**.



[3]

53. MAR 17/P22/Q3/c(ii,iv,v)

(c) But-1-ene reacts with steam as shown to form a mixture of two structural isomers, **S** and **T**.



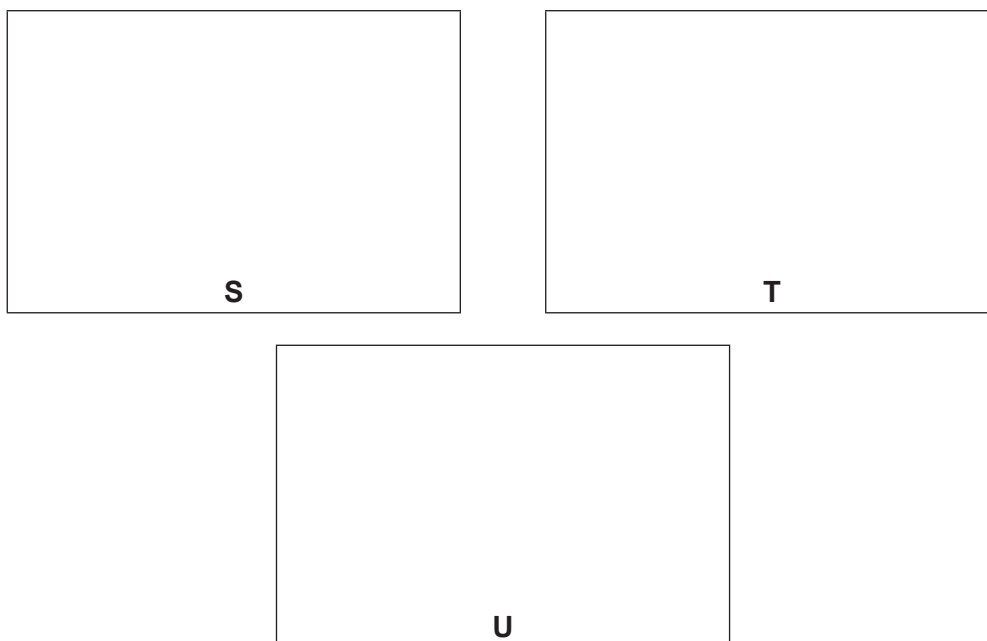
S can be oxidised with acidified potassium dichromate(VI) to form compound **U**.

S and **U** both react with alkaline aqueous iodine.

(ii) State what can be deduced about the structure of **S** from its reaction with alkaline aqueous iodine.

[1]

(iv) Draw the **skeletal** formulae of **S**, **T** and **U**.



[3]

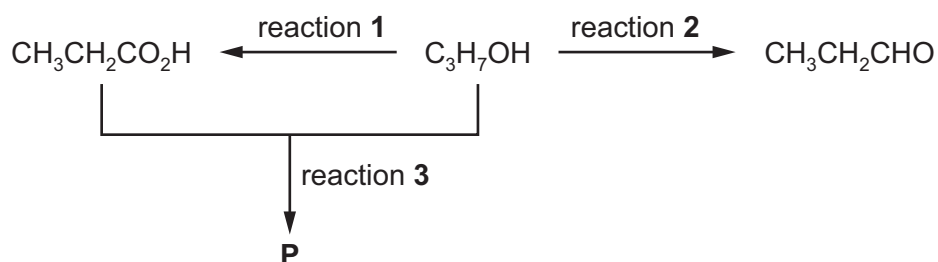
(v) Write an equation to represent the oxidation of **S** to **U** by acidified potassium dichromate(VI).

You should use [O] to represent the oxidising agent.

[1]

54. O/N 16/P21/Q5/b,c

5 A sequence of reactions is shown starting with an alcohol, $\text{C}_3\text{H}_7\text{OH}$.



(b) State the reagents and conditions needed for reaction 1.

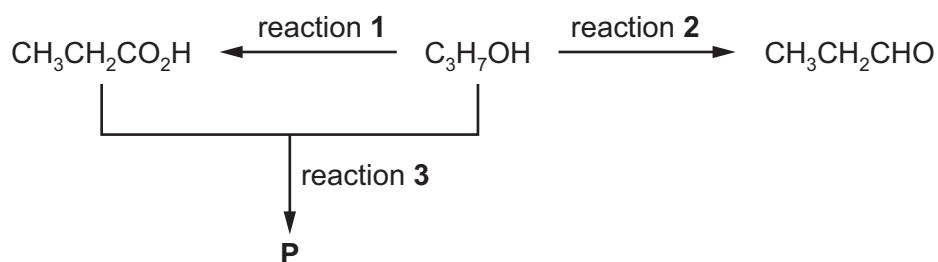
[2]

(c) State the reagents and conditions needed for reaction 2.

[2]

55. O/N 16/P23/Q5/b,c

5 A sequence of reactions is shown starting with an alcohol, $\text{C}_3\text{H}_7\text{OH}$.



(b) State the reagents and conditions needed for reaction 1.

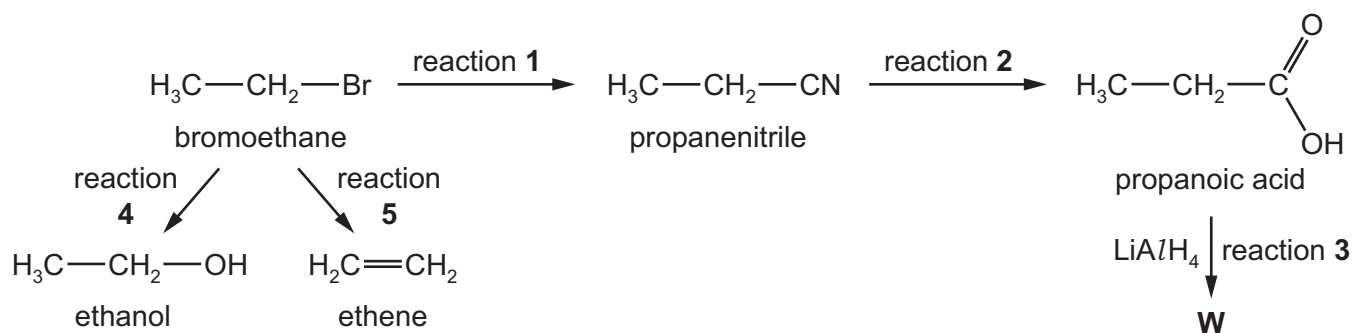
[2]

(c) State the reagents and conditions needed for reaction 2.

[2]

56. M/J 16/P21/Q5/c

5 A reaction sequence is shown.



(c) (i) Reactions **4** and **5** use the same reagent.

Give the reagent and conditions needed for reaction **4**.

reagent

conditions

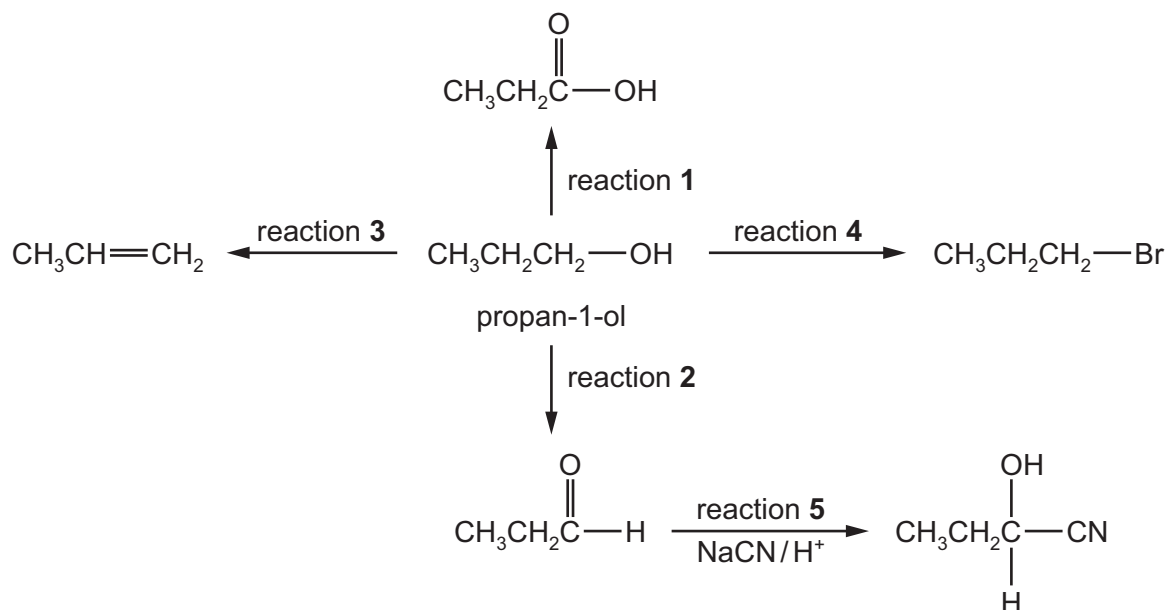
[2]

(ii) Give the conditions needed for reaction **5**.

[1]

57. M/J 16/P22/Q5/a,b

5 A reaction sequence based on propan-1-ol is shown.



(a) Reactions **1** and **2** can both be carried out using the same reagents.

(i) Identify suitable reagents for reactions **1** and **2**.

[1]

(ii) State and explain how the reaction should be carried out to ensure that reaction **2** rather than reaction **1** occurs.

[2]

(b) Identify the necessary reagents and conditions for each of reactions **3** and **4**.

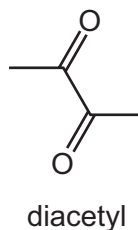
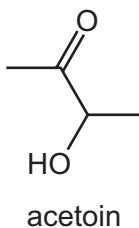
reaction **3**

reaction **4**

[2]

58. M/J 16/P23/Q4/b

4 Acetoin, $\text{CH}_3\text{COCH}(\text{OH})\text{CH}_3$, and diacetyl, $\text{CH}_3\text{COCOCH}_3$, are two of the compounds that give butter its characteristic flavour. Their skeletal formulae are shown.

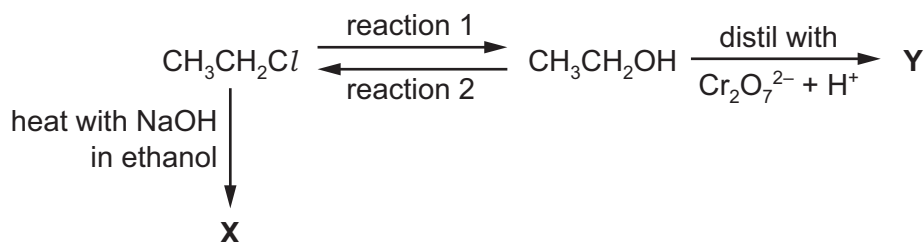


(b) Identify the reagents and conditions necessary for the conversion of acetoin into diacetyl.

[2]

59. O/N 15/P21/Q4/a(ii),c

4 Some reactions involving ethanol are shown.



(ii) State the reagent and conditions required for reaction 1.

[2]

(c) (i) Identify the organic product **Y** which is distilled out of the reaction mixture.

[1]

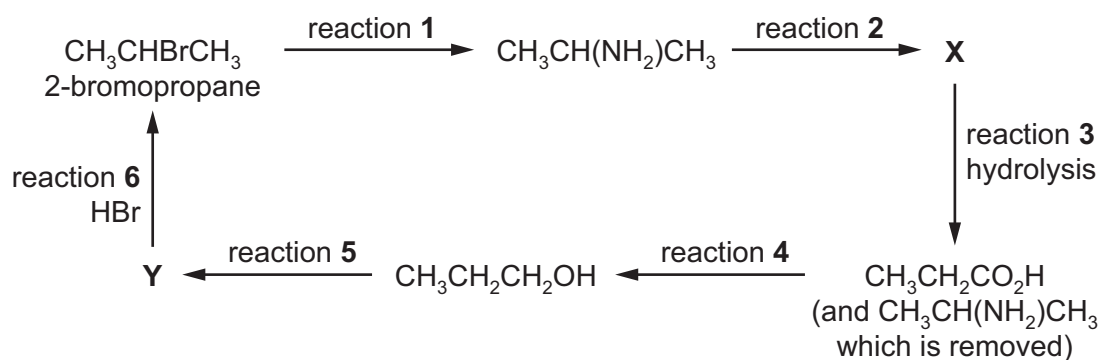
(ii) Explain, in terms of the properties of and intermolecular forces in $\text{CH}_3\text{CH}_2\text{OH}$ and **Y**, why the chosen conditions for the reaction ensure that **Y** is the product.

[3]

60. O/N 15/P22/Q4/b(v)

4 Halogenoalkanes are useful intermediates in the synthesis of a wide variety of compounds.

(b) Some reactions involving 2-bromopropane are shown.

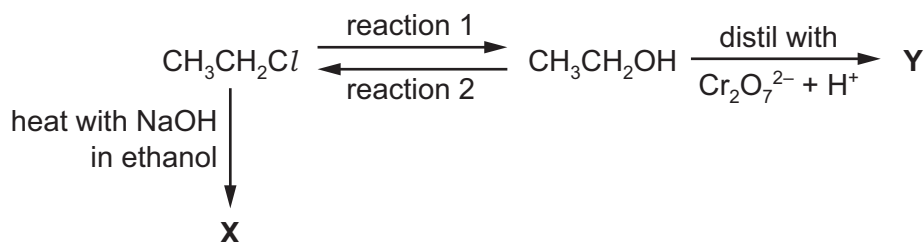


(v) State the name of a solid catalyst for reaction 5.

[1]

61. O/N 15/P23/Q4/a(ii),c

4 Some reactions involving ethanol are shown.



(ii) State the reagent and conditions required for reaction 1.

[2]

(c) (i) Identify the organic product **Y** which is distilled out of the reaction mixture.

[1]

(ii) Explain, in terms of the properties of and intermolecular forces in $\text{CH}_3\text{CH}_2\text{OH}$ and **Y**, why the chosen conditions for the reaction ensure that **Y** is the product.

[3]

62. M/J 15/P21/Q4/b(i)

4 There are four alcohols, **A**, **B**, **C** and **D**, which are structural isomers with the molecular formula $\text{C}_4\text{H}_{10}\text{O}$.

Alcohol **A** does not react with acidified potassium dichromate(VI) solution but **B**, **C** and **D** do.

All four alcohols react with hot, concentrated sulfuric acid to form products with the molecular formula C_4H_8 . **A**, **C** and **D** each give a single product in this reaction. **B** gives a mixture of two structural isomers, one of which shows stereoisomerism.

(b) (i) Give the names of the two structural isomers produced by the reaction of **B** with hot, concentrated sulfuric acid

[2]

63. M/J 15/P23/Q3/b(ii)

3 **A**, **B**, **C**, **D**, **E** and **F** are all structural isomers with the molecular formula $\text{C}_4\text{H}_8\text{O}$.

(b) **D**, **E** and **F** all decolourise bromine and effervesce slowly with sodium metal.

E shows geometrical isomerism. Only **D** has a branched chain.

None of these isomers contains an oxygen atom bonded to a carbon atom involved in π bonding.

None of these isomers contains a chiral centre.

(ii) Identify the gas produced during the reaction of each of these isomers with sodium metal.

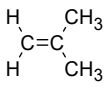
[1]

Answers Section

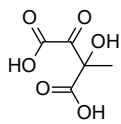
1. M/J 26/P21/Q4/a(i),b(i,iii)

4(a)(i)	(colour of mixture at start of reaction)orange AND (colour of mixture after the reaction)green	1	A any shade of orange A any shade of green I ppt(e) BOD blue / grey for green BOD brown for green (as dichromate is in excess)								
4(b)(i)	<table><tr><th>reagent</th><th>observation with A</th></tr><tr><td>Na(s)</td><td>effervescence / fizzing / bubbling</td></tr><tr><td>2,4-dinitrophenylhydrazine (2,4-DNPH)</td><td>none</td></tr><tr><td>Fehling's reagent</td><td>none</td></tr></table>	reagent	observation with A	Na(s)	effervescence / fizzing / bubbling	2,4-dinitrophenylhydrazine (2,4-DNPH)	none	Fehling's reagent	none	2	M1 I gas produced M1 I name of gas produced M1 I test for any gases M1 I solid disappears/ solid dissolves / colourless solution / flame M2 A (remains) orange solution / remains orange (solution) / no precipitate / no reaction / no observation M2 I 'turns orange'
reagent	observation with A										
Na(s)	effervescence / fizzing / bubbling										
2,4-dinitrophenylhydrazine (2,4-DNPH)	none										
Fehling's reagent	none										
4(b)(iii)	HO₂C-C(=O)-C(=O)-CH₂-CO₂H ✓✓ OR If 2 marks not awarded then max 1 for 2 of the following: • 5 carbon unbranched linear chain • 2 terminal carboxylic acid groups • 2 adjacent ketone groups only •✓	2	A any unambiguous structure I connectivity								

2. M/J 26/P22/Q4/a,b(i)

4(a)(i)	PCl_5 OR SOCl_2 OR HCl OR PCl_3 OR equivalent names	1	A thionyl chloride or sulfur oxide dichloride for SOCl_2 A phosphorus pentachloride or phosphorus trichloride BOD SOCl_2 (aq) A conc. HCl / concentrated hydrochloric acid A hydrogen chloride / HCl (g) I hydrochloric acid alone / HCl (aq) / dilute HCl I NaCl / KCl and conc. H_2SO_4 / conc. H_3PO_4 R PCl_5 (aq)
4(a)(ii)	catalyst OR dehydrating agent	1	A causes dehydration A speeds up / increases rate of reaction and remains unchanged owtte A lowers activation energy / E_A I heterogenous / homogenous I acid R redox
4(a)(iii)	$\text{H}_2\text{C}=\text{C}(\text{CH}_3)_2$ OR  OR 	1	A any unambiguous structure I C_4H_8
4(b)(i)	colour of mixture at start of reactionorange AND colour of mixture after the reactiongreen	1	A any shade of orange A any shade of green BOD blue / grey for green BOD brown for green (as dichromate is in excess)

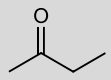
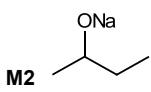
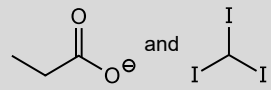
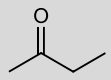
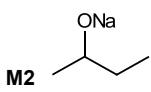
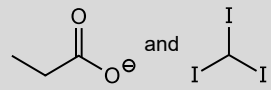
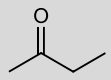
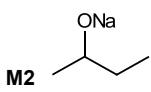
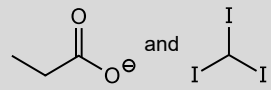
3. M/J 26/P23/Q4/a(i,ii),b,c(iii),d(i)

4(a)(i)	<table><tr><td></td><td>primary hydroxy groups</td><td>secondary hydroxy groups</td><td>tertiary hydroxy groups</td></tr><tr><td>number of groups in one molecule of V</td><td>2</td><td>1</td><td>1</td></tr></table>		primary hydroxy groups	secondary hydroxy groups	tertiary hydroxy groups	number of groups in one molecule of V	2	1	1	2	•✓✓
	primary hydroxy groups	secondary hydroxy groups	tertiary hydroxy groups								
number of groups in one molecule of V	2	1	1								
4(a)(ii)	<table><tr><td>reagent</td><td>observation with V</td></tr><tr><td>Na(s)</td><td>M1 effervescence / fizzing / bubbling</td></tr><tr><td>2,4-dinitrophenylhydrazine (2,4-DNPH)</td><td>M2 none</td></tr><tr><td>Fehling's reagent</td><td>none</td></tr></table>	reagent	observation with V	Na(s)	M1 effervescence / fizzing / bubbling	2,4-dinitrophenylhydrazine (2,4-DNPH)	M2 none	Fehling's reagent	none	2	M1 I gas produced M1 I identity of gas M1 I test for gas M1 I solid disappears / dissolves OR colourless solution M1 I solid made M2 A (remains) orange (solution) / no precipitate / no reaction / no observation I turns orange
reagent	observation with V										
Na(s)	M1 effervescence / fizzing / bubbling										
2,4-dinitrophenylhydrazine (2,4-DNPH)	M2 none										
Fehling's reagent	none										
4(b)(i)	(colour of mixture at start of the reaction)orange (colour of mixture after the reaction)green	1	A any shade of orange at start A after reaction any shade of green BOD blue / grey for green BOD brown for green (dichromate is in excess)								
4(b)(ii)	 OR (HOOC)C=OC(COOH)(OH)CH ₃	1	A any unambiguous structure I connectivity								
4(c)(iii)	methanoic acid	1	A formic acid								
4(d)(i)	(isomer) Y	1	A the structure of isomer Y								

4. M/J 26/P24/Q4/a,b

4(a)	tertiary / 3° (alcohol)		1	I names
4(b)	reagent	observations	3	M1 I Na reacts / gets smaller M2 I antiseptic smell M3 precipitate = ppt = solid = powder = crystals
	Na	M1 Bubbles / fizzing / effervescence		
	acidified K ₂ Cr ₂ O ₇	M2 (Orange to) green (solution)		
	alkaline I ₂ (aq)	M3 Yellow precipitate		

5. MAR 26/P22/Q3/e

3(e)	<table> <tr> <th>reagent and conditions</th><th>observation with butan-2-ol</th><th>organic product(s)</th></tr> <tr> <td>acidified K₂Cr₂O₇</td><td>M1 (orange solution) turns green</td><td></td></tr> <tr> <td>Na(s)</td><td>effervescence</td><td>M2 </td></tr> <tr> <td>M3 alkaline I₂(aq)</td><td>yellow precipitate</td><td></td></tr> </table>	reagent and conditions	observation with butan-2-ol	organic product(s)	acidified K ₂ Cr ₂ O ₇	M1 (orange solution) turns green		Na(s)	effervescence	M2 	M3 alkaline I ₂ (aq)	yellow precipitate		3
reagent and conditions	observation with butan-2-ol	organic product(s)												
acidified K ₂ Cr ₂ O ₇	M1 (orange solution) turns green													
Na(s)	effervescence	M2 												
M3 alkaline I ₂ (aq)	yellow precipitate													

6. O/N 25/P21/Q4/a

4(a)(i)	reduction	1
4(a)(ii)	$\text{PCl}_5 / \text{PCl}_3 / \text{SOCl}_2$ OR KCl / NaCl AND conc. $\text{H}_2\text{SO}_4 / \text{conc. H}_3\text{PO}_4$	1
4(a)(iii)	M1 $\text{C}_2\text{H}_5\text{CH}_2\text{Cl} + \text{NaOH} \rightarrow \text{C}_3\text{H}_6 + \text{H}_2\text{O} + \text{NaCl}$ M2 ethanol (solvent) AND heat / reflux	2
4(a)(iv)	conc. H_2SO_4 OR conc. H_3PO_4 OR Al_2O_3 AND heat	1

7. O/N 25/P22/Q4/c

4(c)(i)	orange to green	1
4(c)(ii)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + 2[\text{O}] \rightarrow \text{CH}_3\text{CH}_2\text{COOH} + \text{H}_2\text{O}$	1

8. O/N 25/P23/Q4/a

4(a)(i)	reduction	1
4(a)(ii)	$\text{PCl}_5 / \text{PCl}_3 / \text{SOCl}_2$ OR KCl / NaCl AND conc. $\text{H}_2\text{SO}_4 / \text{conc. H}_3\text{PO}_4$	1
4(a)(iii)	M1 $\text{C}_2\text{H}_5\text{CH}_2\text{Cl} + \text{NaOH} \rightarrow \text{C}_3\text{H}_6 + \text{H}_2\text{O} + \text{NaCl}$ M2 ethanol (solvent) AND heat / reflux	2
4(a)(iv)	conc. H_2SO_4 OR conc. H_3PO_4 OR Al_2O_3 AND heat	1

9. O/N 25/P24/Q2/c(ii)

2(c)(ii)	$2\text{CH}_3\text{OH} \rightarrow \text{CH}_3\text{OCH}_3 + \text{H}_2\text{O}$	1
----------	--	---

10. O/N 25/P24/Q3/a(i,ii,iii),b(i)

3(a)(i)		1
3(a)(ii)	yellow precipitate	1
3(a)(iii)	elimination	1
3(b)(i)	$(\text{CH}_3)_2\text{CHOH} + \text{Na} \rightarrow (\text{CH}_3)_2\text{CHO}^-(\text{Na}^+) + \frac{1}{2}\text{H}_2$	1

11. M/J 25/P21/Q5/c(ii),d(ii),e(iii)

5(c)(ii)	C is not as strong an acid as B	1
5(d)(ii)	oxidation	1
5(e)(iii)	1,3-dichloropropane	1

12. M/J 25/P21/Q6/c(iii)

6(c)(iii)	$\text{CH}_3\text{CH}(\text{OH})\text{-R}$ where R is represented as H or any alkyl group	1
-----------	---	---

13. M/J 25/P22/Q4/b(i),c(ii,iii),d

4(b)(i)	substitution	1
4(c)(ii)	hydrogen / H_2	1

4(c)(iii)	(It / Na) reduces A / B / C / the alcohol (group) OR (it / Na behaves as) reducing agent OR (it / Na) donates electrons to A / B / C / the alcohol (group)	1
4(d)(i)	(CH ₃) ₃ COH	1
4(d)(ii)	butan-1-ol OR (2) methylpropan-1-ol	1
4(d)(iii)	butan-2-ol OR CH ₃ CH ₂ CH(OH)CH ₃ .	1

14. O/N 24/P22/Q2/b(iv)

2(b)(iv)	alkyl groups are electron donating / have positive inductive effect.	1
	strengthens the O—H bond / makes H ⁺ less likely to be donated	1

15. M/J 24/P21/Q6/c(iii)

6(c)(iii)	ROH + Na → RONa + 1 / 2H ₂ (0.002 mol Q produced 0.002 mol H ₂ gas so) 2 OH groups M1 answer indicates that OH group(s) in Q react with Na to produce the H ₂ in the ratio 1 mol OH : ½ mol H ₂ M2 uses data to show 2OH groups	2
-----------	--	----------

16. O/N 23/P21/Q3/e(i,ii)

3(e)(i)	1 4 0	1
3(e)(ii)	Solution turns from purple to colourless	1
	Orange / red precipitate forms	1
	no reaction	1

17. O/N 23/P22/Q3/c(ii)

3(c)(ii)		1
----------	---	----------

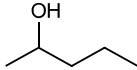
18. O/N 23/P22/Q4/b(ii)

4(b)(ii)	CH ₃ CH(OH)COOH + [O] → CH ₃ COCOOH + H ₂ O	1
----------	--	----------

19. O/N 23/P23/Q3/e(i,ii)

3(e)(i)	1 4 0	1
3(e)(ii)	Solution turns from purple to colourless	1
	Orange / red precipitate forms	1
	no reaction	1

20. M/J 23/P21/Q4/b(i),c

4(b)(i)	cold dilute acidified KMnO ₄	1
4(c)	 M1 molecule with 5C atoms and 1 alcohol group M2 molecule with —CH(CH ₃)OH	2

21. M/J 23/P22/Q3/b

3(b)	acid (conditions) / H ⁺ (aq) AND heat under reflux	1
------	--	----------

22. M/J 23/P23/Q6/b(iii),f(i)

6(b)(iii)	heat + conc. sulfuric acid / conc. phosphoric acid OR heat + Al_2O_3	1
6(f)(i)	substitution	1

23. MAR 23/P22/Q4/a(iii),b(ii)

4(a)(iii)	$\text{C}_4\text{H}_{10}\text{O} / \text{CH}_3\text{CH}(\text{OH})\text{C}_2\text{H}_5 \rightarrow \text{C}_4\text{H}_8 / \text{CH}_3\text{CHCHCH}_3 + \text{H}_2\text{O}$	1
4(b)(ii)	CHI_3 / iodoform / tr(i)iodomethane	1
	$\text{CH}_3\text{CH}_2\text{CO}_2^-$ / propanoate	1

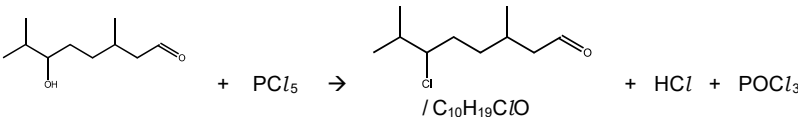
24. O/N 22/P21/Q3/c

3(c)(i)	substitution	1
3(c)(ii)	$ \begin{array}{ccccc} & \text{Cl} & \text{H} & \text{H} & \\ & & & & \\ \text{H}_3\text{C} & -\text{C}- & \text{C}- & \text{C}- & \text{Cl} \\ & & & & \\ & \text{H} & \text{H} & \text{H} & \end{array} $	1

25. O/N 22/P23/Q3/c

3(c)(i)	substitution	1
3(c)(ii)	$ \begin{array}{ccccc} & \text{Cl} & \text{H} & \text{H} & \\ & & & & \\ \text{H}_3\text{C} & -\text{C}- & \text{C}- & \text{C}- & \text{Cl} \\ & & & & \\ & \text{H} & \text{H} & \text{H} & \end{array} $	1

26. M/J 22/P21/Q3/d(i,ii)

3(d)(i)	M1 Q orange $\rightarrow$ green M2 R orange $\rightarrow$ green M3 $-\text{CHO}$ / aldehyde (in both) (and 2° / secondary alcohol in R) reacts / oxidised	3
3(d)(ii)	 M1 correct formula of organic product M2 correct inorganic products	2

27. M/J 22/P23/Q4/b(iv)

4(b)(iv)	$\text{C}_2\text{H}_5\text{COO}^-$ OR $\text{CH}_3\text{CH}_2\text{CO}_2^-$ or propanoate (an)ion	1
----------	--	----------

28. M/J 22/P23/Q5/a(ii)

5(a)(ii)	M1 unbranched 4C structure AND any number of $-\text{OH}$ in any position M2 0.001 mol H_2 made from 0.001 mol Y AND $\text{R}-\text{OH} + \text{Na} \rightarrow \text{RONa} + \frac{1}{2}\text{H}_2$ OR use of 1 OH (group) $\rightarrow \frac{1}{2}\text{H}_2$ M3 $\text{CH}_3\text{CH}(\text{OH})\text{CH}(\text{OH})\text{CH}_3$	3
----------	--	----------

29. O/N 21/P21/Q4/a(i,ii)

4(a)(i)	potassium/sodium dichromate [(VI)] / $K_2Cr_2O_7$ / $Na_2Cr_2O_7$ acidified AND (heat) under reflux	2
4(a)(ii)	$C_2H_5OH + 2[O] \rightarrow CH_3CO_2H + H_2O$	1

30. O/N 21/P22/Q3/e(v)

3(e)(v)	heat AND concentrated H_2SO_4 / concentrated H_3PO_4 / concentrated sulfuric [(VI)] acid / concentrated phosphoric[(V)] acid	1
---------	---	----------

31. O/N 21/P23/Q4/a(i,ii)

4(a)(i)	potassium/sodium dichromate [(VI)] / $K_2Cr_2O_7$ / $Na_2Cr_2O_7$ acidified AND (heat) under reflux	2
4(a)(ii)	$C_2H_5OH + 2[O] \rightarrow CH_3CO_2H + H_2O$	1

32. M/J 21/P22/Q4/c,d

4(c)	dehydrating agent / cause dehydration		1
4(d)(i)	oxidation		1
4(d)(ii)	functional group	present in Z	1
	aldehyde	✓	
	ketone		
	carboxylic acid		

33. M/J 21/P22/Q5/b(ii)

5(b)(ii)	substitution	1
----------	--------------	----------

34. M/J 21/P23/Q4/b

4(b)	<table border="1"> <thead> <tr> <th>reagent</th><th>observation with Q</th><th>observation with R</th></tr> </thead> <tbody> <tr> <td>2,4-DNPH</td><td>M1 red / orange / yellow AND precipitate / solid / crystals</td><td>M2 red / orange / yellow AND precipitate / solid / crystals</td></tr> <tr> <td>$Na_2CO_3(aq)$</td><td>M3 no visible change</td><td>M4 fizz / effervescence</td></tr> </tbody> </table>	reagent	observation with Q	observation with R	2,4-DNPH	M1 red / orange / yellow AND precipitate / solid / crystals	M2 red / orange / yellow AND precipitate / solid / crystals	$Na_2CO_3(aq)$	M3 no visible change	M4 fizz / effervescence	4
reagent	observation with Q	observation with R									
2,4-DNPH	M1 red / orange / yellow AND precipitate / solid / crystals	M2 red / orange / yellow AND precipitate / solid / crystals									
$Na_2CO_3(aq)$	M3 no visible change	M4 fizz / effervescence									

35. M/J 21/P23/Q5/b(ii)

5(b)(ii)	M1 citric acid has a tertiary alcohol group so is not oxidised	1
	M2 X has a secondary alcohol group which is oxidised (to a ketone)	1

36. MAR 21/P22/Q4/c

4(c)(i)	oxidising agent	1
4(c)(ii)	M1: (excess dichromate and) heat under reflux M2: to allow full oxidation (of alcohol and aldehyde groups)	2

37. O/N 20/P21/Q3/c

3(c)(i)	M1: potassium dichromate[(VI)] M2: acid(ified) AND (heat under) reflux	2
---------	--	----------

3(c)(ii)	(M1: correct identity of R and statement re: reaction 3 ONLY ketone reduced) R (is 2-hydroxybutanoic acid) AND as (only) C=O / ketone reduced (M2: correct explanation re: strength of reducing agents) NaBH₄ cannot reduce the COOH / carboxylic acid OR LiAlH₄ can reduce the COOH / carboxylic acid	2
3(c)(iii)	M1: Presence of :CN (if bonding shown, must be unambiguous triple bond) M2: curly arrow from :CN lone pair to carbonyl carbon M3: correct dipole AND curly arrow from double bond to oxygen M4: correct intermediate drawn	4
3(c)(iv)	$\text{C}_2\text{H}_5\text{CH}(\text{OH})\text{CN} + \text{HCl} + 2\text{H}_2\text{O} \rightarrow \text{C}_2\text{H}_5\text{CH}(\text{OH})\text{COOH} + \text{NH}_4\text{Cl}$	1
3(c)(v)	Any two of three absorption references: - absorption 2200–2250 (cm⁻¹) shows presence of C≡N - lack of absorption at 1680–1730 (cm⁻¹) shows lack of C=O - lack of absorption at 2500–3000 (cm⁻¹) shows lack of RCO₂–H / O–H in RCO₂H	2

38. O/N 20/P21/Q4/b(i)

4(b)(i)	HI(g) / PI ₃ / P and I ₂	1
---------	--	---

39. O/N 20/P22/Q4/a(i,ii,iii,v)

4(a)(i)	oxidation	1
4(a)(ii)	M1: potassium dichromate[(VI)] M2: acid(ified) AND (heat under) reflux	2
4(a)(iii)	structure of H OR CHO(CH₂)₂CH₂O⁻ Na⁺	1
4(a)(v)	esterification	1

40. O/N 20/P23/Q3/c

3(c)(i)	M1: potassium dichromate[(VI)] M2: acid(ified) AND (heat under) reflux	2
3(c)(ii)	(M1: correct identity of R and statement re: reaction 3 ONLY ketone reduced) R (is 2-hydroxybutanoic acid) AND as (only) C=O / ketone reduced (M2: correct explanation re: strength of reducing agents) NaBH₄ cannot reduce the COOH / carboxylic acid OR LiAlH₄ can reduce the COOH / carboxylic acid	2
3(c)(iii)	M1: Presence of :CN (if bonding shown, must be unambiguous triple bond) M2: curly arrow from :CN lone pair to carbonyl carbon M3: correct dipole AND curly arrow from double bond to oxygen M4: correct intermediate drawn	4
3(c)(iv)	$\text{C}_2\text{H}_5\text{CH}(\text{OH})\text{CN} + \text{HCl} + 2\text{H}_2\text{O} \rightarrow \text{C}_2\text{H}_5\text{CH}(\text{OH})\text{COOH} + \text{NH}_4\text{Cl}$	1

3(c)(v)	Any two of three absorption references: - absorption 2200–2250 (cm⁻¹) shows presence of C≡N - lack of absorption at 1680–1730 (cm⁻¹) shows lack of C=O - lack of absorption at 2500–3000 (cm⁻¹) shows lack of RCO₂H / O–H in RCO₂H	2
---------	---	----------

41. O/N 20/P23/Q4/b(i)

4(b)(i)	HI(g) / PI ₃ / P and I ₂	1
---------	--	----------

42. MAR 20/P22/Q2/e(ii,iii)

2(e)(ii)	C ₂ H ₅ CH(OH)C ₂ H ₅ + HCl → C ₂ H ₅ CH(Cl)C ₂ H ₅ + H ₂ O	1
2(e)(iii)	$ \begin{array}{c} \text{OH} \\ \\ \text{H}_3\text{C}-\text{C}-\text{CH}_2\text{CH}_3 \\ \\ \text{CH}_3 \end{array} $	1

43. MAR 20/P22/Q3/a(i)

3(a)(i)	M1 acidified / H ⁺ Cr ₂ O ₇ ²⁻ / (potassium / sodium) dichromate OR manganate(VII) / MnO ₄ ⁻ / KMnO ₄ M2 (heat under) reflux	2
---------	---	----------

44. O/N 19/P22/Q3/c(i,ii)

3(c)(i)	C ₂ H ₅ OH + PCl ₅ → C ₂ H ₅ Cl + POCl ₃ + HCl	1
3(c)(ii)	substitution	1

45. O/N 19/P22/Q4/b

4(b)(i)	$ \begin{array}{c} \text{H}_3\text{C} \quad \text{H} \\ \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \\ \text{HO} \quad \text{CH}_2\text{OH} \end{array} $	1
4(b)(ii)	(CH ₃) ₂ CHCH(OH)CH ₂ OH	1
4(b)(iii)	optical (isomerism)	1
4(b)(iv)	C ₅ H ₁₂ O ₂ + 3[O] → C ₅ H ₈ O ₃ + 2H ₂ O	1

46. M/J 19/P21/Q4/c(iii)

4(c)(iii)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
-----------	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

47. M/J 19/P23/Q6/b

6(b)(ii)	V = primary / 1° alcohol	1
	W = tertiary / 3° alcohol	1
6(b)(ii)	CH ₃ (CH ₂) ₂ CH ₂ OH + Na → CH ₃ (CH ₂) ₂ CH ₂ ONa + ½ H ₂	1

6(b)(iii)	M1 Reagent	$\text{H}^+ / \text{Cr}_2\text{O}_7^{2-}$	3
	M2 Observations for V	orange to green	
	M3 Observations for W	no change / remains orange	

48. MAR 19/P22/Q4/b(iv)

4(b)(iv)	SOCl_2 OR PCl_5 OR PCl_3 OR $\text{c}(\text{oncentrated}) \text{HCl}$	1
----------	--	----------

49. O/N 18/P22/Q4/a(iii,iv)

4(a)(iii)	$\begin{array}{c} \text{OH} \\ \\ \text{H}_3\text{C}-\text{C}-\text{H} \\ \\ \text{CO}_2\text{H} \end{array} + [\text{O}] \rightarrow \begin{array}{c} \text{O} \\ \\ \text{H}_3\text{C}-\text{C}-\text{CO}_2\text{H} \end{array} + \text{H}_2\text{O}$	1
4(a)(iv)	$\text{Na}_2\text{Cr}_2\text{O}_7 / \text{K}_2\text{Cr}_2\text{O}_7$ AND (dilute) $\text{H}_2\text{SO}_4 / \text{H}^+(\text{aq})$ / acidified	1

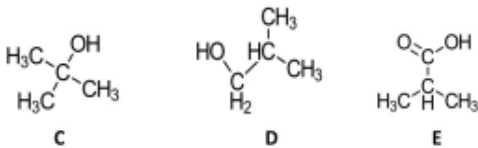
50. M/J 18/P23/Q4/a(ii)

4(a)(ii)	ketone	1
----------	--------	----------

51. O/N 17/P22/Q4/a

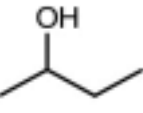
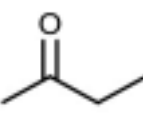
4(a)		<u>c</u> oncentrated $\text{H}_2\text{SO}_4 / \text{H}_3\text{PO}_4$ AND NaBr	5
	1	OR (red) P / Br_2 OR HBr	
	2	aqueous / dilute NaOH / KOH	
	3	<u>c</u> oncentrated $\text{H}_2\text{SO}_4 / \text{H}_3\text{PO}_4$ OR $\text{Al}_2\text{O}_3 / \text{P}_4\text{O}_{10}$ / pumice / porous pot / SiO_2	
	4	(ethanolic) HBr	
4 marks for column 1 (one per row) 1 mark for col 2			

52. M/J 17/P22/Q4/b(i)

4(b)(i)	C = (2-)methylpropan-2-ol / $(\text{CH}_3)_3\text{COH}$ / any unambiguous structure	1
	D = (2-)methylpropan-1-ol / $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$ / any unambiguous structure	1
	E = (2-)methylpropanoic acid / $(\text{CH}_3)_2\text{CHCO}_2\text{H}$ / any unambiguous structure	1
	 C D E	

53. MAR 17/P22/Q3/c(ii,iv,v)

3(c)(ii)	S has CH_3CHOH OR methyl / CH_3 group next to CHOH	1
----------	--	----------

3(c)(iv)	S = 	1
	T = 	1
	U = 	1
3(c)(v)	$\text{CH}_3\text{CHOHCH}_2\text{CH}_3 + [\text{O}] \rightarrow \text{CH}_3\text{COCH}_2\text{CH}_3 + \text{H}_2\text{O}$	1

54. O/N 16/P21/Q5/b,c

5(b)	$\text{H}^+ / \text{Cr}_2\text{O}_7^{2-}$ (heat under) reflux	1 1	2
5(c)	$\text{H}^+ / \text{Cr}_2\text{O}_7^{2-}$ (heat and) distil	1 1	2

55. O/N 16/P23/Q5/b,c

5(b)	$\text{H}^+ / \text{Cr}_2\text{O}_7^{2-}$ (heat under) reflux	1 1	2
5(c)	$\text{H}^+ / \text{Cr}_2\text{O}_7^{2-}$ (heat and) distil	1 1	2

56. M/J 16/P21/Q5/c

(c) (i)	sodium/potassium hydroxide aqueous	[1] [1]	[2]
(ii)	ethanol	[1]	[1]

57. M/J 16/P22/Q5/a,b

5 (a) (i)	acidified / H^+ AND potassium / sodium dichromate	[1]	[1]
(ii)	distillation (rather than reflux) (ensures aldehyde escapes) to avoid further oxidation / to avoid forming acid / as reflux causes further oxidation	[1] [1]	[2]
(b)	reaction 3 – (conc) H_2SO_4 / (conc) H_3PO_4 or Al_2O_3 / pumice / porcelain / porous pot / ceramic AND heat reaction 4 – KBr / NaBr with (conc) H_2SO_4 or (red)P and $\text{Br}_2 / \text{PBr}_3$ AND heat	[1] [1]	[2]

58. M/J 16/P23/Q4/b

(b)	$\text{H}_2 / \text{Cr}_2\text{O}_7^{2-}$ or names heat / reflux / warm	[1] [1]	[2]
-----	--	------------	-----

59. O/N 15/P21/Q4/a(ii),c

(ii)	NaOH/KOH warm/heat/reflux AND aqueous	[1] [1]	[2]
(c) (i)	CH ₃ CHO/ethanal	[1]	[1]
(ii)	CH ₃ CH ₂ OH higher bpt than CH ₃ CHO ora due to hydrogen bonding in ethanol/stronger IMFs prevents further oxidation owtte	[1] [1] [1]	[3]

60. O/N 15/P22/Q4/b(v)

(v)	aluminium oxide	[1]	[1]
-----	-----------------	-----	-----

61. O/N 15/P23/Q4/a(ii),c

(ii)	NaOH/KOH warm/heat/reflux AND aqueous	[1] [1]	[2]
(c) (i)	CH ₃ CHO/ethanal	[1]	[1]
(ii)	CH ₃ CH ₂ OH higher bpt than CH ₃ CHO ora due to hydrogen bonding in ethanol/stronger IMFs prevents further oxidation owtte	[1] [1] [1]	[3]

62. M/J 15/P21/Q4/b(i)

(b) (i)	but-1-ene/ 1-butene but-2-ene/ 2-butene	[1] [1]	[2]
---------	--	------------	-----

63. M/J 15/P23/Q3/b(ii)

(ii)	Hydrogen	[1]	[1]
------	----------	-----	-----